

Perceptual glimpses are locally accumulated and globally maintained at distinct processing levels

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eLife Assessment

This study presents **valuable** findings on the physiological and computational underpinnings of the accumulation of intermittent glimpses of sensory evidence. While the authors present **solid** evidence to support their claims, a more exhaustive characterisation of how the different signals interact could further strengthen their case. The work will be of interest to cognitive and systems neuroscientists working on decision-making

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Abstract

Making decisions often requires the integration of multiple pieces of information. An extensive body of research has investigated the neural architecture underpinning evidence accumulation in perceptual tasks where information is continuously present, but less is known about how this neural architecture operates in situations affording only intermittent glimpses of an evidence source. In two electroencephalography (EEG) experiments, participants judged the direction of up to two pulses of motion evidence separated by gaps of varying duration. Our behavioural analysis found that participants used both pulses but underutilised the second, and showed no systematic decrease in accuracy as a function of gap duration. At the neural level, motor beta lateralisation tracked cumulative evidence across pulses, maintaining a sustained representation of the decision variable through the gap and until response. In contrast, the centroparietal positivity (CPP), a previously-characterised signature of evidence accumulation, built up transiently to a peak that scaled with each pulse's contribution to the decision variable (i.e. the absolute belief update it produced), falling back to baseline in between pulses. These patterns were recapitulated in a model where pulse-information transiently integrated at the CPP level is fed to and maintained at a bounded motor level, so that information presented later on in a trial is integrated only if a bound has not yet been reached.

1. Introduction

Perceptual decision-making is widely thought to involve a process of evidence integration over time (Hyafil et al., 2023). An extensive body of research has successfully linked these integration computations to multiple neural decision signals in various species (Gold & Shadlen, 2007 [↗](#); Hanks & Summerfield, 2017 [↗](#); Heekeren et al., 2008 [↗](#); O’Connell & Kelly, 2021 [↗](#); Shadlen & Kiani, 2013 [↗](#)) revealing a neural decision architecture that includes both effector-selective motor preparation signals, and motor-independent signals that appear to reflect integration operations at an intermediate processing level (O’Connell & Kelly, 2021 [↗](#); Okazawa & Kiani, 2023 [↗](#)). Such neural decision processes have mainly been studied in the context of tasks requiring immediate judgements about a single continuously-presented stimulus. However, in many real-life situations a decision-maker is afforded only a short glimpse of a relevant evidence source and it is uncertain whether and when they will receive an additional glimpse, as, for example, when evaluating an object among moving traffic that may be intermittently occluded. This scenario has begun to be studied using an intermittent- evidence or “gaps” task (Kiani et al., 2013 [↗](#)), in which up to two short pulses of noisy evidence for a certain motion direction are presented separated by a variable temporal gap, with a behavioural response required after a delayed cue. Previous studies using this task have found that both evidence pulses are utilised, but not fully or equally (Azizi & Ebrahimpour, 2023 [↗](#); Kiani et al., 2013 [↗](#)), and although a general information leakage account could be ruled out based on the absence of any decline in accuracy with gap duration, the neural underpinnings of the task, including the functional roles of effector-selective and motor-independent processes in handling and maintaining the intermittent evidence, and how such information underutilisation can arise, have remained unclear.

In humans, two neurophysiological signatures of decision formation can be utilised alongside behavioural data to gain insight into mechanisms underlying perceptual tasks (O’Connell & Kelly, 2021 [↗](#)). First, spectral activity in the beta band (13-30 Hz), long observed to decrease over motor cortex with preparation of a contralateral limb (Pfurtscheller & Lopes da Silva, 1999; Stancák & Pfurtscheller, 1995 [↗](#)), has been found in decision tasks to build at an evidence-dependent rate, reaching a threshold level contralateral to, and coincident with, the executed action (Geuzebroek et al., 2023 [↗](#); Kelly et al., 2021 [↗](#); O’Connell et al., 2012 [↗](#); Steinemann et al., 2018 [↗](#)). Second, a centroparietal positivity (CPP) in the human electroencephalogram (EEG) has been found to exhibit similar evidence-dependent buildup dynamics in conventional perceptual tasks with continuous stimuli such as random dot motion, but with an earlier latency; it begins building at a stimulus-aligned onset time, rises at a rate that scales with evidence strength, and peaks at the time of action choice when immediate execution is allowed (Kelly & O’Connell, 2013 [↗](#); O’Connell et al., 2012 [↗](#); Steinemann et al., 2018 [↗](#)). In contrast with motor preparation signals, the CPP exhibits evidence-integration dynamics even when no motor output is required (O’Connell et al., 2012 [↗](#)). In delayed-response tasks where actions are deferred until some time after the perceptual choice is made, the CPP peaks at decision commitment and falls back to its baseline level, whereas relative motor preparation reflected in motor beta lateralisation (MBL) sustains until action execution (Twomey et al., 2016 [↗](#)). These findings are consistent with the CPP tracking an intermediary, motor-independent evidence integration process that is continuously fed to downstream motor preparation where a sustained representation of the DV is maintained.

More recently, a few studies have highlighted that the CPP can behave differently in expanded judgement tasks, where evidence is presented as a series of discrete tokens that need to be integrated rather than as a continuous, noisy stream. In these tasks, centroparietal signals exhibit transient token-evoked positivities that scale with the immediate evidence, rather than tracking the evolution of an ongoing decision variable (DV) across tokens (Parés-Pujolràs et al., 2025 [↗](#); Wyart et al., 2012 [↗](#), 2015 [↗](#)). Motor Beta Lateralisation (MBL), meanwhile, tracks the evolving decision variable across discrete tokens over multiple seconds, by stepping up or down according

to the evidence in each token and sustaining in between them (Murphy et al., 2021 [↗](#); Parés-Pujolràs et al., 2025 [↗](#)). These findings again point to the operation of an intermediary decision process - one that can be flexibly configured to compute different quantities depending on the nature of the task.

The Gaps task, like expanded judgment tasks, involves temporally disparate pieces of information, but it holds key distinctions that demand dedicated investigation. In expanded judgement tasks, tokens typically bear little or no perceptual noise, but the overlap of the underlying generative distributions means they often provide conflicting information. The nature of the stimuli thus encourages appraisal of each token as a distinct object, potentially explaining the transient pattern of CPP activity. In contrast, the Gaps task (Kiani et al., 2013 [↗](#)) involves pulses of evidence that are perceptually noisy but always indicate a consistent motion direction, thus appearing as glimpses of the same object rather than a series of individual objects. Only one study has examined neural signatures of decision formation during the Gaps task: using EEG, Azizi & Ebrahimpour (2023) [↗](#) showed that the CPP exhibits an evoked positivity after each pulse that appears to decay during the gap, suggesting a local processing role. However, how these pulse-evoked activities relate to the computations underpinning an ongoing decision, interact with motor preparation to determine choice, and relate to any information underutilisation, has not been examined.

In this study, we recorded EEG during the Gaps task (Kiani et al. 2013 [↗](#)) in two experiments with slightly different settings for the variable temporal gaps to ensure generality of findings. Our primary goal was to characterise the dynamics of the CPP and MBL in the context of this intermittent evidence task, and functionally relate them to behavioural indices of integration across tokens. We found that participants based their choices on both pulses but with a weaker influence of pulse 2, and coherence-dependent weightings on choice were mirrored in the magnitude of transient pulse-evoked CPPs. These patterns could be explained by a decision architecture where the CPP conducts two separate rounds of evidence accumulation, feeding the cumulative evidence within each pulse to a downstream motor process indexed by MBL. MBL in turn tracks a continuous representation of the state of the DV, maintaining the accumulated information through the gap and until response. Our behavioural and modelling results indicated that participants engaged in a bounded accumulation process, so that pulse information was only integrated while the DV encoded in MBL had not reached the bound. Additionally, given the dynamic structure of the task involving evidence onsets and offsets, we explored the dynamics of attention, as indexed by occipital alpha signals, and these were suggestive of a key role in filtering irrelevant evidence, and in modulating the weight of pulses on choice.

2. Results

2.1. Behavioural results

In this study, participants engaged in a motion direction discrimination task where, on most trials, two pulses of partially-coherent dot motion (indicated by yellow dot colour) were separated by variable temporal ‘gaps’ in which motion was random, and thus uninformative (indicated by blue dot colour; **Fig. 1A** [↗](#)). Though coherence could differ, the direction was always consistent between the two pulses, so that they would be construed as two glimpses of the same evidence source. Our two experiments (Exp. 1: N = 22; Exp. 2: N = 21) differed only in the delays separating the two pulses. In both experiments, coherence levels were titrated for each participant using a staircase procedure aiming for an average 70% accuracy following a single evidence pulse. High (Exp. 1: M = 44.73%, SD = 12.64%; Exp 2: M = 33.04%, SD = 13.41%) and low (Exp. 1: M = 26.75%, SD = 7.74%; Exp 2: M = 19.82%, SD = 8.04%) coherences were then set to be, respectively, 25% higher and 25% lower than the staircase- estimated coherence (see *Methods*), and small adjustments were made between blocks if performance substantially changed (*Figure S1*).

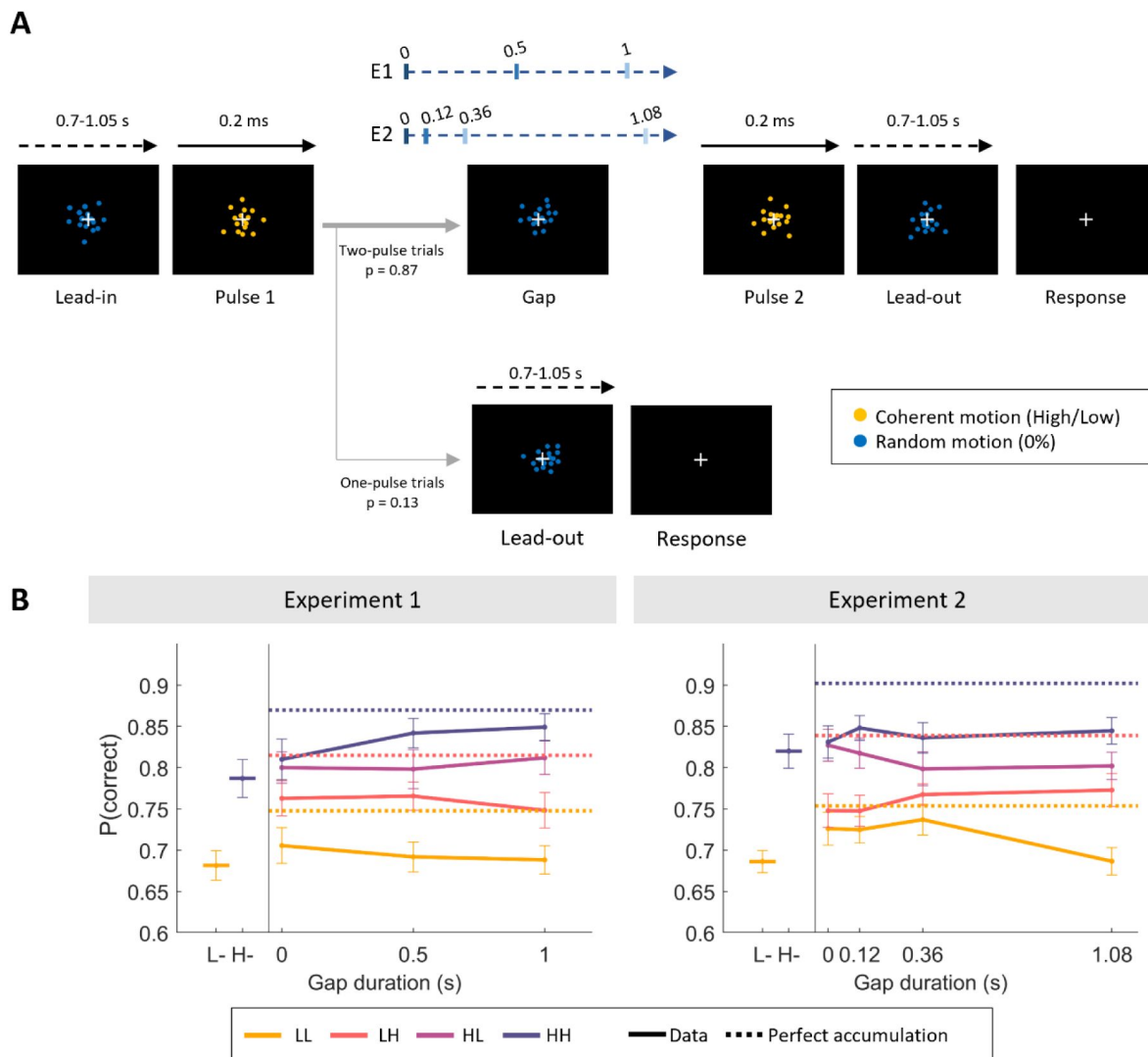


Figure 1.

Task and behavioural results.

A. Experimental structure. Participants viewed a random dot stimulus that coherently moved either to the left or right during yellow evidence pulses, of which there was sometimes one, but usually two, separated by a variable gap in evidence indicated by blue dot colour. **B.** Grand-averaged (mean $\pm$ SEM) accuracy in two-pulse trials, sorted by pulse coherence (low, L/ high, H), order (LL, LH, HL, HH) and gap duration. Data points in the left-hand section indicate single-pulse accuracies. Overlaid horizontal dashed lines indicate the expected accuracy of a perfect accumulator, based on the grand-average performance on one-pulse trials. Note that a perfect accumulator predicts identical accuracies for the HL and LH conditions, and therefore the two lines overlap.

As expected, participants' accuracy scaled with motion coherence and number of pulses. In one-pulse trials, participants were more accurate in the high coherence compared to the low coherence trials (*Eq. 1*, Exp 1: $\beta_1 = 0.33 \pm 0.05$, $p < 0.001$; Exp 2: $\beta_1 = 0.42 \pm 0.06$, $p < 0.001$, **Figure 1A**). There was a further benefit in accuracy from the second pulse when present, indicated by sensitivity to coherence in both the first (*Eq. 2*, Exp 1: $\beta_1 = 0.27 \pm 0.02$, $p < 0.001$; Exp 2: $\beta_1 = 0.30 \pm 0.03$, $p < 0.001$) and second (*Eq. 2*, Exp 1: $\beta_2 = 0.18 \pm 0.02$, $p < 0.001$; Exp 2: $\beta_2 = 0.06 \pm 0.02$, $p < 0.001$) pulses. When both pulses in a two-pulse trial were the same coherence, participants performed better compared to one-pulse trials, exhibited both in high and low coherence trials in Experiment 2 (*Eq. 3*, Exp. 2: $\beta_1 = 0.19 \pm 0.06$, $p = 0.003$; $\beta_3 = 0.04 \pm 0.05$, $p = 0.39$), but only significantly so in high coherence trials in Experiment 1 (*Eq. 3*, Exp. 1: $\beta_1 = 0.21 \pm 0.06$, $p = 0.001$; $\beta_3 = 0.18 \pm 0.03$, $p < 0.001$). However, they did not perform as well as expected from a perfect accumulator (**Figure 1B**), estimated using performance in one-pulse trials to predict expected accuracy in two-pulse ones (see *Eq. 5-7*; see *Methods*).

The fact that the regression coefficients for the second pulse were smaller than those estimated for the first one indicates that participants' choices were guided more strongly by the first piece of evidence they received. On average, accuracies were lowest for the most difficult trials (both pulses had low motion coherence, LL), intermediate for trials with mixed coherences (LH, HL) and highest for the easiest trials (both pulses had high motion coherence, HH). However, we also observed a significant pulse-order effect: when the two presented pulses had different coherences, participants were more accurate when the first pulse was the high-coherence one (HL) compared to trials where the first pulse was low-coherence (LH) in both experiments (*Eq. 4*, Exp 1: $\beta_2 = -0.11 \pm 0.03$, $p < 0.001$; Exp 2: $\beta_2 = -0.12 \pm 0.02$, $p < 0.001$; **Figure 1B**). This is in line with participants' choices being more strongly influenced by the first evidence pulse.

Importantly, there was no main effect of gap duration on accuracies (*Eq. 2*, Exp 1: $\beta_3 = 0.05 \pm 0.02$, $p = 0.22$; Exp 2: $\beta_3 = -0.03 \pm 0.05$, $p = 0.51$; **Figure 1A**), suggesting an absence of leakage of information from the first pulse, in line with previous work (Azizi & Ebrahimpour, 2023; Kiani et al., 2013). However, gap duration did interact with evidence strength in different ways. In Exp 1 participants' accuracy increased with gap duration when the first pulse coherence was high, and decreased with gap duration when it was low (*Eq. 2*, Exp 1: β_4 (p1 x gap interaction) = 0.10 ± 0.02 , $p < 0.001$). In Exp 2, accuracies increased with gap duration when the second pulse coherence was high, and decreased when it was low (*Eq. 2*, Exp 2: β_5 (p2 x gap interaction) = 0.12 ± 0.05 , $p = 0.008$). Since these interactions were inconsistent across experiments and we did not have prior hypotheses regarding such effects, we leave these to one side and focus on the main effects of coherence. In what follows, we combine EEG and computational modelling to test the possibility that the greater sensitivity to the first pulse arises from the participants engaging in a bounded accumulation process where decisions are sometimes terminated early, curtailing the extent to which the second pulse benefits accuracy and driving the coherence order effects.

2.2. EEG results

2.2.1. Activity during the gap

We then turned to neural correlates of evidence accumulation during the task, specifically, MBL and the CPP. Across the two experiments and in all gap conditions, MBL showed evidence-dependent buildup continuously through the sequence of pulses with a temporal lag (**Fig. 2A**), sustaining at a coherence-dependent level (see *Fig. S2*; *S3*) throughout the gap between the two pulses and until response. In contrast, the CPP exhibited a transient build-up following each evidence pulse, falling back to baseline levels prior to the second pulse (**Fig. 2B**).

Given that our task included strong external (colour) cues indicating time windows with and without evidence, we further investigated the dynamics of attentional engagement with the stimulus, as indexed by occipital alpha activity (Foxe & Snyder, 2011; Van Diepen et al., 2019),

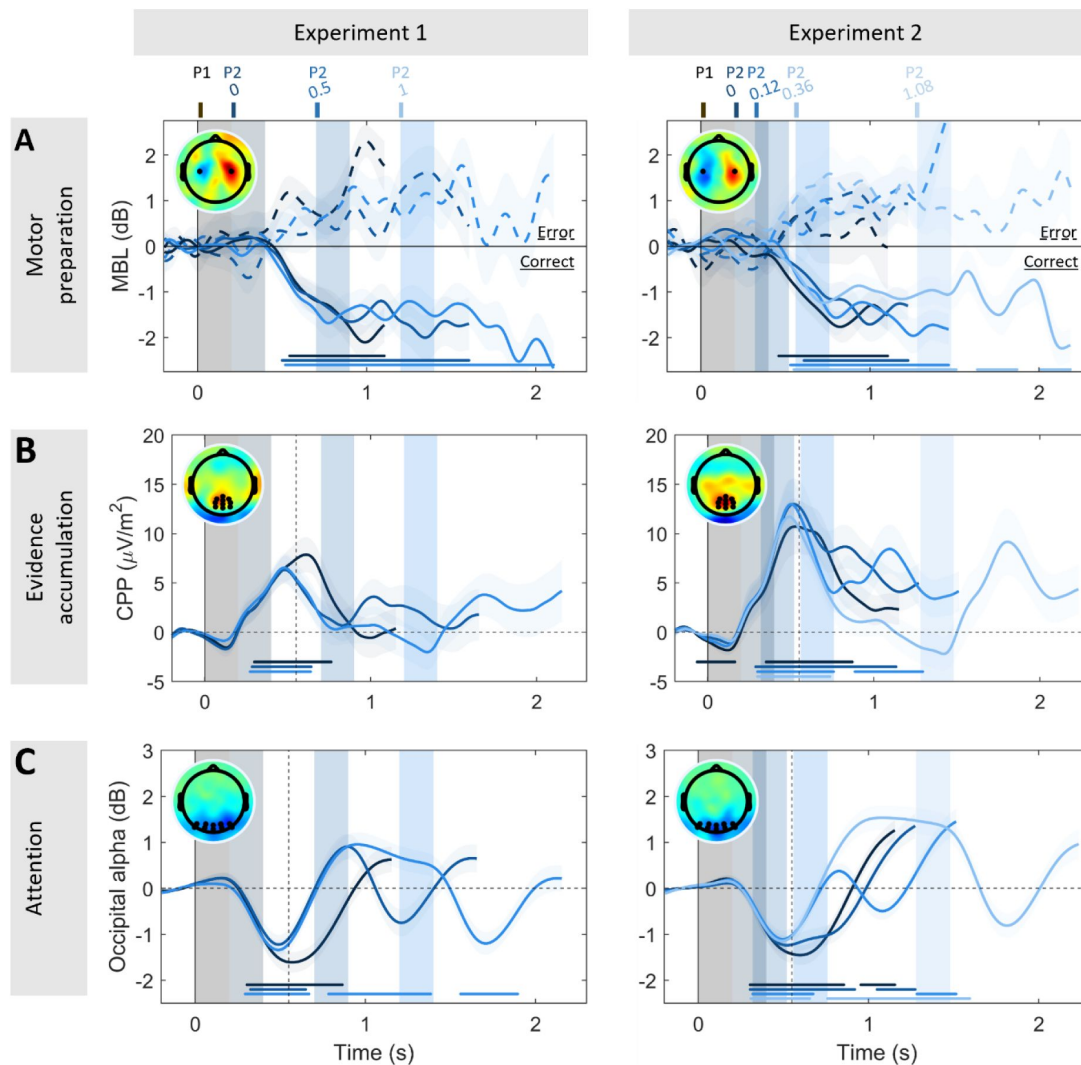


Figure 2.

Neural correlates of evidence accumulation and motor preparation across temporal gaps.

A. Grand-averaged ($\pm$ SEM) motor beta lateralisation (MBL) [Contra- Ipsilateral hemispheres with respect to correct response hand], sorted by accuracy (*solid line* correct, *dashed line* error). Participants started preparing a motor response in response to the first pulse, and MBL was sustained throughout the gap and until the time of behavioural response across all gap conditions in both experiments. **B.** Grand-averaged ($\pm$ SEM) centroparietal positivity (CPP) across all temporal gap conditions. The CPP showed transient evoked responses following each pulse. Note that the evoked amplitudes by P1 (CPP-P1) were systematically higher than those evoked by P2 (CPP-P2) (Exp. 1: CPP-P2 = 52% of CPP-P1 peak amplitude; Exp. 2: CPP-P2 = 58% of CPP-P1 peak amplitude; see Fig. S4). **C.** Grand-averaged ($\pm$ SEM) occipital alpha power for each of the gap conditions, baseline-corrected before the onset of the first pulse and collapsed across coherence. Occipital alpha power showed a steep decrease following pulse onset (which was clearly indicated by a change in the colour of the moving dots), and steep increase overshooting the baseline level once the dots changed colour again to indicate non-coherent. In all panels, shaded areas indicate the timing of evidence pulses for the various gap durations, which are colour coded (see top axis). Topographies illustrate MBL before response (right minus left responses, **A**) in correct trials, and CPP/alpha power 550 ms post stimulus onset (*dashed line*, **B,C**). Further, markers along the bottom indicate times where the signal significantly differed from zero (two-tailed cluster- based permutation test, $p < 0.005$) in only correct trials for MBL for consistency of polarity (**A**), and for all trials pooled in the case of the CPP (**B**) and alpha (**C**). Note that evidence-dependent buildup dynamics are clearly delayed in MBL relative to the CPP, and this aligns with previous research supporting the contention that the CPP lies upstream from motor preparation (Steinemann et al. 2018); such time delays were not a focus of the present study because in such delayed-response conditions we do not have access to the time of decision commitment.

throughout trial time. In both experiments, we observed strong occipital alpha desynchronisation (i.e. a decrease relative to baseline) during the periods in which the CPP indicated active evaluation of the yellow-dot evidence periods (**Fig. 2C**), and strongly elevated alpha above baseline levels in between those pulse-processing periods. Adding to the above behavioural finding of no main effect of gap duration, this suggests that the participants successfully gated stimulus evaluation according to dot colour, selectively attending to the temporal windows in which decision-relevant information is available for evaluation, and ignoring the decision-irrelevant gap periods.

2.2.2. Centroparietal signals track multiple rounds of evidence accumulation during a single decision

Next, we turned to coherence-dependent effects. The CPP evoked after the first pulse (CPP-P1) scaled with the strength of evidence as expected, reaching higher amplitudes after high coherence pulses compared to low-coherence ones over a cluster of centroparietal electrodes (Exp. 1: $p = 0.009$, Exp. 2: $p = 0.031$; two-tailed cluster-based permutation test; **Fig. 3A-D**). Furthermore, single-trial variability in CPP-P1 amplitude significantly modulated the first pulse's weight on choice: in trials where evoked CPP-P1 amplitudes were particularly high, the coherence of pulse 1 had a stronger influence on behaviour (*Fig. S11A*).

The coherence of the second pulse modulated peak amplitude in the standard way, with larger CPP amplitudes in trials with high pulse 2 coherence, although this effect was only significant in Experiment 2 (Exp. 1: $p = 0.165$, Exp. 2: $p = 0.009$; two-tailed cluster-based permutation test; **Fig. 3F,H**). CPP-P2 amplitude was also influenced by the coherence of Pulse 1: on trials where the first pulse was low coherence, CPPs evoked by the second pulse reached higher amplitudes (Exp. 1: $p = 0.018$, Exp. 2: $p = 0.005$; two-tailed cluster-based permutation test; see **Fig. 3F,H**). Additionally, CPP-P2 amplitude was generally smaller than CPP-P1 amplitude, which was the case whether we computed CPP-P2 amplitudes after baseline-correction before pulse 2 onset (Exp. 1: $t(21) = 2.29$, $p = 0.032$; Exp. 2: $t(20) = 4.19$, $p < 0.001$) or after correction for CPP-P1 overlap (Exp. 1: $t(21) = 1.93$, $p = 0.067$; Exp. 2: $t(20) = 3.25$, $p = 0.003$; see *Methods* for details).

2.2.4. A single bounded decision process can account for key behavioural and neural patterns

We then aimed to identify a computational model that could account for the key behavioural and CPP amplitude effects observed in our data. Similar to previous studies ([Azizi & Ebrahimpour, 2023](#); [Kiani et al., 2013](#)) there was no evidence of leak in our data, given that no general decrease in accuracy was observed for longer gap durations. The lack of a main effect of gap duration on accuracies also indicates that participants did not accumulate the incoherent motion presented during the gaps. Instead, the primacy effect observed in our behavioural results, with higher accuracies in HL than LH trials, is consistent with a bounded process where participants sometimes terminated evidence accumulation early, failing to use all available evidence. Thus, we fitted a single bounded accumulator model with three free parameters (bound, high- and low-coherence drift rates), where evidence accumulation can only proceed during the pulse time windows and is terminated upon bound crossing (see *Methods* for full details). Given our behavioural observations, we assumed 1) no accumulation during the gap intervals, and 2) no leak, both of which would predict a generalised decrease in accuracy over longer gap durations that was not empirically observed. Our model therefore only allowed evidence accumulation during the pulse time windows, and assumed the DV was sustained at the value attained at the end of pulse 1 throughout the gap. If the bound was not crossed by the end of the trial, the sign of the DV at the end of the trial determined the model's predicted choice.

This simple single-DV model could account for the order effects observed in behavioural data, where the first pulse had a greater bearing on choice than the second one, while at the same time capturing the generally greater accuracy of two-pulse trials compared to single-pulse ones (**Fig.**

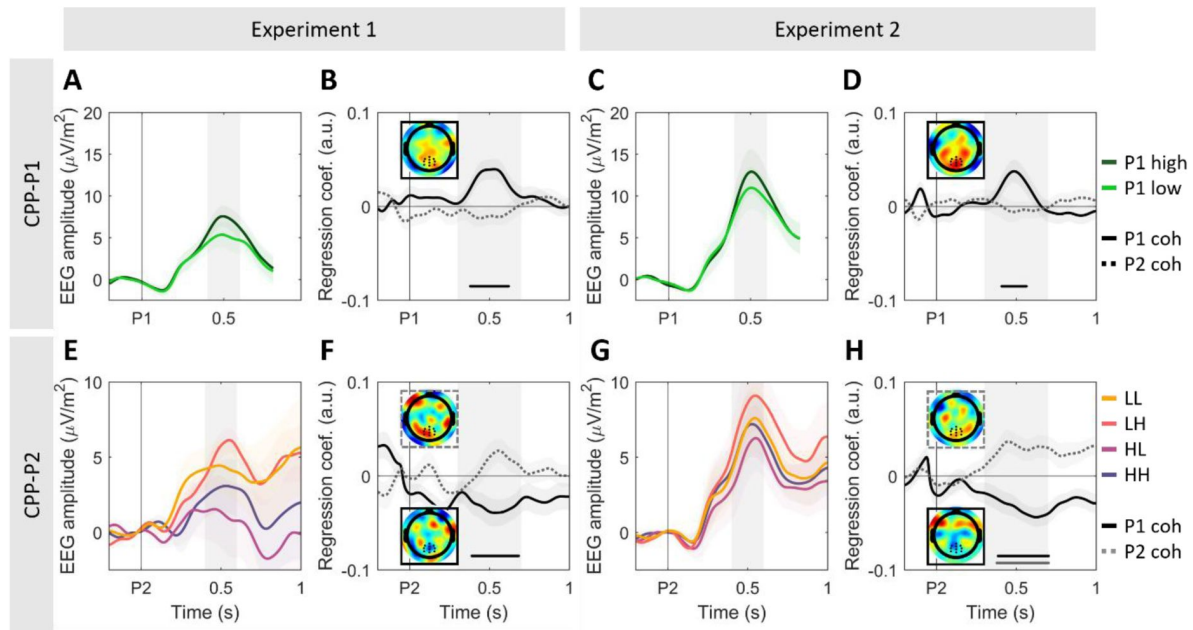


Figure 3.

CPP amplitude following each pulse scales with its coherence, and the strength of evidence in pulse 1 additionally inversely modulates amplitudes following pulse 2.

A,C. Grand-averaged ($\pm$ SEM) centroparietal activity after pulse 1 (CPP-P1), averaged across all gap conditions. **B,D.** Standardized regression coefficients illustrating the effect of P1 (solid line) and P2 (dashed line) coherence on evoked CPP-P1 amplitudes. Inset topographies illustrate the coherence effect [P1 high - P1 low] at [0.4-0.6s] after P1. **E,G.** Grand-averaged ($\pm$ SEM) centroparietal activity after pulse 2 (CPP-P2), averaged over all non-zero gaps in the two experiments. For illustration purposes, we subtracted the average activity of long gap trials with the same P1 coherence from the epoched EEG traces, between [-0.2 to 1.4s] from P1 presentation, and baseline corrected traces [-100 to 0ms] before P2. This isolates the neural activity uniquely associated with the presentation of the second pulse in short gap trials (<1s), where potentials evoked by the first and second pulses overlapped, by removing the activity associated with the first pulse from the P2-evoked traces, including baseline trends (see Fig. S4). The qualitative patterns did not vary if this correction was not applied (see Fig S5). Note also the difference in the y axis scale compared to panels A,C. **F,H.** Standardized regression coefficients (mean $\pm$ SEM) illustrating the effect of P1 (solid line) and P2 (dashed line) coherence on evoked CPP-P2 amplitudes. Inset topographies illustrate the difference between [P1 high - P1 low], solid outline; [P2 high - P2 low], dashed outline) [0.4-0.6s] after P2. The negative effect of P1 illustrates higher amplitudes for P1-low trials. *Markers along the bottom of panels B,D,F and H indicate significant clusters where coherence effects differed from zero (two-tailed cluster-based permutation test, $p < 0.05$). Shaded areas indicate the tested time window. See Fig. S3 for full-trial ERP traces, sorted by gap and coherence conditions.

4A [↗](#)). In this model, this partial benefit of the second pulse arises from the fact that the DV is further influenced by it only on the trials where the DV has not already reached a bound. As expected, a greater proportion of trials terminate early in the P1-high trials compared to P1-low trials (**Fig. 4B** [↗](#)). Note that this simple diffusion model has no way to produce systematic gap-dependent effects, meaning that predicted accuracies are forced to be flat across gaps; again, since the observed gap-dependent effects were unexpected and differed in our two experiments, we did not pursue further modelling approaches to attempt to capture them in the current study (but see *Fig. S9, S10 and S11* for exploratory analyses on how temporal expectations might influence integration processes and evidence weighting).

Then, we simulated the neural activity corresponding to MBL and the CPP (**Fig. 5** [↗](#)). Our empirical results suggested that motor preparation as indexed by MBL tracks the state of the bounded decision variable in a sustained manner, emerging shortly after the first pulse and sustaining until response. Thus, we simulated MBL by taking the signed values of the model-derived DV throughout the trial and averaging across trials. These exhibited a similar pattern to the empirical data, with the signed DV showing opposite-sign patterns for error vs. correct trials (**Fig. 5A,D** [↗](#), *top*; cf. **Fig. 2A** [↗](#)). If a bound was hit, the DV was set to stay stable in our model, and therefore recapitulated the way in which motor preparation responses here (**Fig. 2A** [↗](#)) and in previous studies (e.g. [Murphy et al., 2021](#) [↗](#); [Parés-Pujolràs et al., 2025](#) [↗](#); [Twomey et al., 2016](#) [↗](#)) sustain until response execution. Note that, in our simulations (**Fig. 5A,D** [↗](#)), the average traces do not always reach the estimated bound: this is because, in a fraction of trials, the bound was never hit, bringing averaged values away from the bound value.

In turn, our results suggested that the transient CPP activity reflected local, pulse-derived updates to the DV (i.e. $|\Delta DV|$), not its global state. Thus, we simulated CPP activity by taking the absolute values of the cumulative evidence gathered within each pulse ($|\Delta DV|$) on single trials (in keeping with the always-positive nature of the CPP; see [Afacan-Seref et al., 2018](#) [↗](#); [Kelly et al., 2021](#) [↗](#)) and averaging across trials. If a bound was hit by the overall DV, the CPP traces were set to return linearly to 0 within 200 ms, as empirically observed in action-locked traces in self-paced tasks (e.g. [Steinemann et al., 2018](#) [↗](#)), and it did not resume upon presentation of a second pulse. Given that the CPP traces in our grand-averaged data (**Fig. 2B** [↗](#)) fell fully back to baseline by the end of the longer gaps, CPP traces were also assumed to linearly return to zero as soon as the dots turned blue at the end of evidence pulses if a bound had not been hit earlier on. Such a pattern assumes that the CPP generators are active during periods where evidence is presented (yellow dot pulses), but become inactive and fall down to baseline as soon as evidence stops (blue dot periods). During pulse processing, the CPP feeds its output to the downstream motor level, which sustains its current level during the gap rather than deactivating. In this way, the motor process continuously represents the overall DV throughout the trial: the evidence that had been integrated by the CPP during pulse 1 (ΔDV_{P1}) is maintained at the motor level through the gap, is potentially updated by another round of accumulation by the CPP during pulse 2 provided the bound was not already reached, and thereafter further sustains up until response execution. Given the absence of previous empirical data on how signals may fall down without having hit a bound, we assumed the same decay rate in this case as when a bound is hit (linear return to zero within 200 ms).

The simulated single-pulse accumulation traces from our model captured the sensitivity of both CPP-P1 and CPP-P2 to motion coherence. The difference in CPP-P1 slopes for high vs. low coherence pulses is simply explained by the high vs. low drift-rate associated with these conditions (**Fig. 5B,E** [↗](#)). In the more complex case of CPP-P2, the evoked activity after P2 reached higher amplitudes in trials where P1 was low coherence (**Fig. 5C,F** [↗](#)). Our model can account for this due to two effects: First, because P1-Low trials are less likely to terminate early (**Fig. 4B** [↗](#)), and therefore fewer trials take a value of zero during P2. Second, because among those trials that do not terminate early, the level reached by the DV by the end of pulse one (ΔDV_{P1}) is on average lower for low- than high-coherence (**Fig. 4C** [↗](#); **Fig. 5B,E** [↗](#)), and therefore more evidence from P2 needs to be integrated before the bound is hit and accumulation terminated. Thus, the higher CPP-

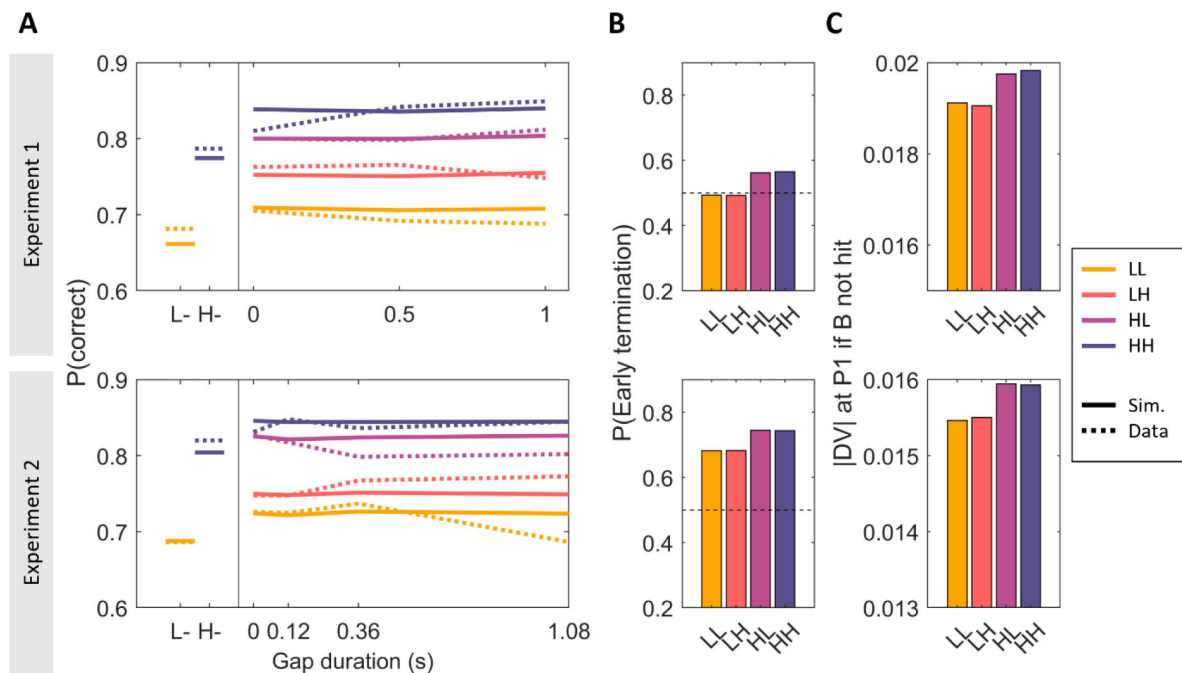


Figure 4.

Simulated behavioural data

($N = 50000$) of a simple bounded accumulator model with three parameters (B (bound), $d1$ (high drift rate), $d2$ (low drift rate)), fit to all data points for each experiment separately. **A.** Grand-averaged simulated (solid lines) accuracy in single- (left) and two-pulse (right) trials, sorted by pulse coherence and gap duration in experiments 1 (top) and 2 (bottom). Observed data are overlaid (dotted) for comparison. Simulated data recapitulated the observed order effects. For simplicity, the model was not endowed with any means to account for effects of gap duration. **B.** Proportion of simulated two-pulse trials that hit a bound during the first pulse, and therefore did not use the second pulse evidence to make a choice. **C.** Absolute value of the decision variable (DV) at the end of pulse 1 in the subset of trials where the bound was not hit during that pulse, sorted by condition.

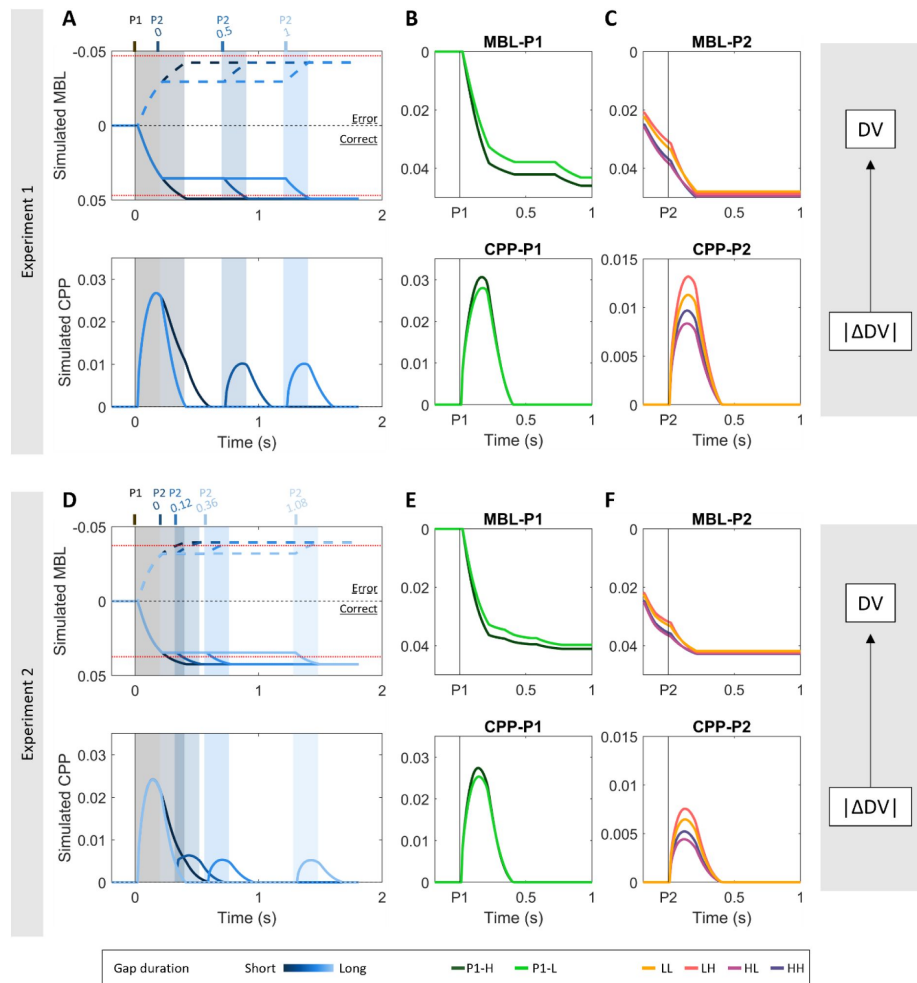


Figure 5.

Model-derived simulations of motor beta lateralisation (MBL) and centroparietal positivity (CPP) recapitulate empirical data.

A,D. Simulated MBL (*top*) and CPP (*bottom*) traces over the whole trial, sorted by gap duration. MBL traces are additionally sorted by accuracy (solid line = correct; dashed lines = error). The simulations recapitulate the bimodal shape of the CPP across the whole trial, with the second pulse reaching lower amplitudes than the 1st one in both experiments (see **Fig. 2B**). In turn, by assuming that MBL signals track the state of the DV throughout the trial and because no leak or decay is assumed, our simulations could also recapitulate the sustained nature of motor lateralisation signals throughout the gap. Note that DVs were simulated so that positive signs are associated with correct choices, and negative signs to errors. Dashed red lines indicate the bounds for error/correct trials, respectively. **B,E.** Simulated MBL-P1 (*top*) and CPP-P1 (*bottom*), sorted by P1 coherence, and averaged across all gap durations and accuracies. Note that the step-like changes in MBL traces are due to averaging the DV values over the various gap durations. No such effect is observed in simulated CPP potentials because traces are derived from P1 activity exclusively. **C,F.** Simulated MBL-P2 (*top*) and CPP-P2, sorted by P1 and P2 coherence, and averaged across all gap durations and accuracies. Our simulations recapitulate the observed coherence effects at both CPP-P1 (see **Fig. 3A,C**) and CPP-P2 (see **Fig. 3E,G**). Note that, for CPP simulations, we plot the estimated build-up at P2 assuming that activity started at 0 and not considering any potential overlap from earlier on in the trial, for consistency with the empirical traces where CPP-P1 activity is removed from the P2-evoked traces. In contrast, the MBL-P2 traces continue from their post-P1 state, accounting for the sustained nature of the signal (see **Fig. S2,S3** for empirical MBL-P1/P2 traces). Note also that, for the sake of simplicity, our CPP simulations do not assume any delay between stimulus presentation and signal build up. Therefore, our simulated CPPs start building up immediately after pulse onset, while empirical ones start with approximately a 0.2s delay. Similarly, although motor preparation signals lag the motor independent buildup dynamics at the CPP level, we assumed instant transmission here for simplicity. Since only accuracy and not reaction time was to be captured in this delayed-response task, it was not necessary to estimate transmission delays.

P2 amplitudes reached in trials with low P1 coherence reflected both a higher proportion of trials where participants engaged in accumulation of P2, and the fact that they had to cover a longer distance to reach the bound (i.e. larger changes in the DV ($|\Delta DV_{P2}|$) were possible before a bound was hit).

The model also predicted differences in CPP amplitude in correct/incorrect trials, which our dataset broadly recapitulated (Fig. S6). In particular, it predicted stronger CPP-P1 coherence-dependent effects in correct vs. incorrect trials, and it also predicted an interaction between accuracy and P2-coherence on CPP-P2 amplitude in trials with mixed coherences (HL, LH).

Finally, we assessed the extent to which some of our models' assumptions influenced the simulated CPP profile. For example, we assumed for parsimony that the CPP would deactivate with the same temporal dynamics when a decision process terminates because a bound has been hit, as when accumulation is reset at the start of the gap, but we do not know this to be the case. In fact, our simulations show the CPP falling down to baseline in all gap conditions, while empirical data show this pattern for the longest gap condition only. We therefore conducted additional simulations to illustrate how different assumptions about the decay duration (i.e. time to go back to 0) and delay-to-fall (i.e. the amount of time elapsed before the signal stops accumulating and starts going down) would affect the CPP's profile during the gap in our two experiments (Fig. S7). These parameters affected the CPP shape during the gap, but the patterns of coherence-dependent CPP-P1 and CPP-P2 amplitudes remained unchanged. Longer decay durations better recapitulate the observed empirical fall-down in short gap conditions, where signals fail to reach baseline before the second pulse onset. These simulations suggest that our assumption of a 200ms decay time may have underestimated that of the true signal dynamics, but dedicated experiments will be required to precisely quantify the decay characteristics of the CPP in these scenarios. In turn, allowing for some post-decisional evidence accumulation before the signal starts falling down also better captures some features of our empirical data, such as the higher amplitudes and longer peak latencies observed in the zero gap condition (Fig. S7). We also simulated an alternative scenario where the signal is not assumed to decay (Fig. S8), which would be predicted if the CPP's neural generators remained active throughout the gap, encoding a sustained representation of the DV, so long as the bound has not been crossed. These simulations did not recapitulate the CPP's full drop to baseline in the long gap conditions, and produced distinctly smaller CPP-P2 evoked potentials relative to CPP-P1 (Exp. 1: CPP-P2 = 19% of CPP-P1 amplitude; Exp. 2: CPP-P2 = 13% of CPP-P1 amplitude) than observed in the empirical data (see Fig. 2B), which were more closely approximated by the model assuming a full reset during the gap (Exp. 1: CPP-P2 = 53% of CPP-P1 amplitude; Exp. 2: CPP-P2 = 35% of CPP-P1 amplitude; see *Methods* for full details on this analysis). Thus, the data are more consistent with a full reset of the intermediate accumulation process reflected in the CPP at the end of the pulse containing informative evidence, regardless of it having crossed a bound, with the maintenance of intermediate, potentially subthreshold levels of the overall DV being encoded solely at the downstream level of motor preparation.

3. Discussion

A wealth of studies has investigated the neural mechanisms underpinning evidence accumulation in tasks where information is continuously available, identifying multiple neural decision signals at various levels in the sensorimotor hierarchy (Gold & Shadlen, 2007; O'Connell & Kelly, 2021; Okazawa & Kiani, 2023). However, little is known about the mechanisms underpinning decision-making in contexts where evidence from a given source comes in temporally separate bouts. This is important not only because many real-life decisions require this kind of integration, but also because neural decision signals can exhibit functional dissociations when probed in more complex contexts (Okazawa & Kiani, 2023; Parés-Pujolràs et al., 2025). In this study, we asked: how do neural decision signals support evidence accumulation if information availability is paused for a variable amount of time? Focussing on two human EEG decision signals, the

centroparietal positivity (CPP) and motor beta lateralisation (MBL), we show that they reflect processes with distinct roles during decision formation in this context. Across two experiments, we show that the CPP does not exhibit sustained activity during the gap between evidence pulses, but exhibits transient activations following each pulse that track distinct rounds of evidence accumulation, and reflect its absolute contribution to an overall decision variable ($|\Delta DV|$). In contrast, MBL exhibited build-up during pulses and sustained activity during the gap between, consistent with the encoding of a signed DV evolving across pulses. Our behavioural and neural data can be explained by a simple bounded accumulation process, where participants sometimes terminate decisions early despite a cost in accuracy.

Our finding that the CPP exhibits transient activations tracking multiple rounds of evidence accumulation that feed into a single decision provides further insight into its functional role. Most previous studies using continuous evidence paradigms showed that the CPP steadily grows throughout an evidence accumulation process, in parallel with motor preparation, consistent with its reflecting the absolute value of an underlying, continuously- accumulating DV (Kelly & O'Connell, 2013 [DOI](#); O'Connell et al., 2012 [DOI](#); Steinemann et al., 2018 [DOI](#)). In contrast, in our task the CPP exhibits two distinct build-up processes that align with the two pulses presented in each trial. This suggests that the CPP builds up while accumulation is ongoing, and disengages when the accumulation process is paused or a bound is hit. This was further supported by simulations of the CPP profile under various signal decay assumptions, which showed that empirical signals were better recapitulated by a process that assumes signal decay when evidence is paused (*Fig S6*). Furthermore, we observed that the amplitude of each pulse-evoked CPP reflected the weight of that pulse on choice. At the grand average level, the initial pulse systematically evoked higher amplitudes than the second one, consistent with its greater influence on choice behaviour (**Fig. 2B** [DOI](#)). Further, single-trial regressions confirmed that fluctuations in CPP-P1 amplitude for a given coherence level were linked to behavioural variability, with higher-than-average CPPs associated with a stronger weighting of the first pulse on choice (*Fig S11*). A link between CPP amplitude and behavioural weighting was also observed in the one previous study of the CPP during the Gaps task (Azizi & Ebrahimpour, 2023 [DOI](#)): in that study, it was the second pulse that had a stronger weight on choice rather than the first (discussed below), and CPP amplitudes were accordingly larger for pulse 2. Thus, although the direction of the behavioural bias differed across studies, in both, the bias in CPP amplitudes was in the same direction as the behavioural bias.

Our results support the idea that the CPPs following sequential pulses capture multiple rounds of evidence accumulation that are part of a single decision, and that its amplitude reflects each pulse's partial contribution to an evolving DV. In contrast to the transient pattern of activity that characterises the CPP in our task, MBL started shortly after the first pulse, and was sustained at a coherence dependent level throughout the whole trial duration (**Fig 2** [DOI](#) & *Fig S2*). This is in line with multiple previous studies, which have identified sustained encoding of decision variables in motor preparatory signals (Murphy et al., 2021 [DOI](#); Parés-Pujolràs et al., 2025 [DOI](#); Wyart et al., 2012 [DOI](#), 2015 [DOI](#)). Put together, our findings support the idea that in this task the CPP provide belief updates ($|\Delta DV|$) to downstream motor signals that maintain a sustained representation of the evolving DV. Thus, our study identifies new task-dependent functional dissociations between distinct levels of the neural architecture underpinning decision making.

The dynamics within these levels and interactions between them were further illuminated by our computational models. In our two experiments, both the behavioural and neural effects could be reproduced with a two-level bounded accumulation model, in which an intermediate accumulator operates locally on contiguous periods of evidence, feeding its output on to a thresholded motor level where the cumulative total is sustained. Our model assumed that participants stop accumulating evidence if a bound is hit and that accumulation is paused during the gap between pulses; while the cumulative total is held at the motor level, the intermediate accumulator resets. By assuming that the CPP becomes inactive and falls down during the gaps, our simulations could recapitulate key CPP characteristics observed in our task including the opposite effects of the first

and second pulse coherence on its amplitude. The fact that trials with low coherence first pulses were less likely to be terminated early, and had to cover a longer distance-to-bound upon presentation of the second pulse explained why CPP- P2 reached higher amplitudes in those trials.

A key feature of our model which enabled it to account for our behavioural observations was that accumulation was bounded, so that decisions could be terminated before all of the evidence has been sampled and accumulated. Given that the motion in any two sequential pulses in the gaps task always went in the same direction, that behavioural responses were deferred until a later cue, and that single-pulse accuracies were far from ceiling, the optimal strategy would have been to perfectly accumulate all available evidence; by setting a bound, our participants often missed some information and thus achieved suboptimal accuracy. Interestingly, behaviour consistent with such early decision terminations have been observed in other types of task (Balsdon et al., 2020 [↗](#); Kiani et al., 2008 [↗](#)) previous studies using the gaps task found the opposite bias in behavioural weighting, where the second pulse had a stronger influence on choice than the first (Azizi & Ebrahimpour, 2023 [↗](#); Kiani et al., 2013 [↗](#)). Differences in trial structure, difficulties and training may account for this discrepancy. For example, our study had comparatively long lead-in periods before the first pulse (min. 0.7s compared to min. 0.4s of previous studies), which may have helped them to prepare for full integration of the first pulse. We also used a narrower range of coherences than the previous studies, which possibly lends itself to calibrating a bound to achieve acceptable accuracy while saving cognitive effort. Further, even though the probability of a single pulse trial was the same, the single-pulse staircasing procedure we used at the beginning of the task may have encouraged a strategy more focused on the information in the first pulse. Resolving the factors that encourage early termination strategies will ultimately require dedicated experimental manipulations.

Our findings regarding the CPP's contribution to decisions based on temporally separate bouts of evidence aligns with the results of recent work in expanded judgement tasks (Parés-Pujolràs et al., 2025 [↗](#); Wyart et al., 2012 [↗](#), 2015 [↗](#)). In expanded judgement studies, participants view a sequence of discrete, easily-identified tokens that support one of two choices and must integrate over multiple tokens to make an accurate decision. In these tasks, CPP responses to discrete tokens have been shown to scale with context-optimal belief updates in classic non-volatile contexts that prescribe perfect linear accumulation (Wyart et al., 2012 [↗](#), 2015 [↗](#)) as well as in volatile environments where the correct answer may change within a given trial, which prescribe a nonlinear transformation of the token information being integrated (Parés-Pujolràs et al., 2025 [↗](#)). That is, evoked CPP activities track how much a DV changed following each piece of evidence, but not the absolute running sum of accumulated evidence. This stands in contrast to the results from conventional decision-making tasks where evidence is presented continuously, and in the form of high-noise continuous stimuli (Geuzebroek et al., 2023 [↗](#); Gherman et al., 2024 [↗](#); O'Connell et al., 2012 [↗](#); Steinemann et al., 2018 [↗](#)) rather than low- noise discrete tokens. Our task represents a middle ground between these two types of paradigms: we used noisy continuous evidence, but grouped it in two temporally discrete, sequential windows. Thus, participants needed to integrate over time to discriminate the information in each evidence pulse, but also across evidence pulses to maximise their accuracy. This allowed us to show that, in conditions where evidence is noisy and continuous but can be paused, the CPP's profile separately tracks multiple rounds of evidence accumulation. When such updates occur in discrete steps as in expanded judgement tasks, or in temporally discrete bouts as in our gaps task, the CPP does not seem to keep track of the state of an overall DV, but rather it encodes temporally discrete updates. However, it remains unclear to what extent the CPP may behave differently in other contexts. In particular, a common feature between expanded judgement tasks and our design is that the presence of evidence is unambiguously indicated, in our case by colour cues. These colour cues may have acted to trigger initiation and termination of two rounds of evidence accumulation. However, it is plausible that in the absence of such external cues the CPP may have exhibited a sustained pattern of activity consistent with

continuous monitoring of the environment (Geuzebroek et al., 2023 [↗](#)), potentially plateauing during periods with incoherent dot motion that should elicit close-to-zero, or heavily reduced, momentary belief updates on average. This will require a dedicated experiment to test.

Our study also leaves some open questions. Given that the coherence-dependent effects of gap duration on behaviour were not expected and were inconsistent across the two experiments (Fig 1B [↗](#)), we did not aim to capture them in our modelling. However, surmising that they might relate to dynamic expectations arising from the temporal structure of the task, we sought to gain insight into dynamics of attentional engagement and expectation through exploratory analyses of occipital alpha activity (Fig 2C [↗](#), Fig S9). Occipital Alpha-band power has been linked to attention and temporal expectations (Rohenkohl & Nobre, 2011 [↗](#)), and also to the onset (Van Den Brink et al., 2021 [↗](#)) and build-up rate (Kelly & O'Connell, 2013 [↗](#)) of evidence accumulation signals like the CPP. In our two experiments, the gap durations between the two pulses followed different distributions: in Exp 1 gaps were uniformly distributed, whereas in Exp 2 they were skewed towards shorter gaps durations. We found evidence that occipital alpha power was sensitive to this difference (Fig S9), in that it was relatively reduced during the longer gaps in Exp 1, where they were relatively more likely, compared to Exp 2. We also found that phasic alpha decreases additionally modulated the weight of the second pulse on behaviour in Exp 2 (Fig S11), suggesting that phasic alpha activations may influence evidence integration. However, more work is required to specifically test how tonic and phasic occipital Alpha may interact with evidence accumulation processes, or influence behaviour through other mechanisms.

In sum, our study provides novel insight into the neurocomputational mechanisms supporting decision making in a context where decisions need to be made on the basis of a single stimulus feature that is only intermittently available. Our task allowed us to identify novel functional characteristics of a well-established neural decision signal in the human EEG: the CPP. We show that in tasks where perceptually noisy evidence is presented intermittently in short bouts, the CPP does not maintain a sustained representation of cumulative evidence, but rather engages in multiple rounds of evidence accumulation, which in turn update DVs represented in downstream areas. This is consistent with recent work showing that the CPP encodes belief updates rather than the ongoing state of a DV in expanded judgment tasks involving the integration of a series of stimuli that are individually easy to identify (Parés-Pujolràs et al. 2025), and suggests that the CPP may only trace the entire evidence accumulation process underpinning decisions when the physical evidence is presented in one continuous stream (Geuzebroek et al., 2023 [↗](#); Kelly & O'Connell, 2013 [↗](#); O'Connell et al., 2012 [↗](#); Steinemann et al., 2018 [↗](#)). Our results thus add to growing work suggesting that the neural architecture underlying evidence accumulation for decision-making can exhibit flexible functional roles in different contexts, underpinning adaptive behaviour.

4. Methods

Participants: Twenty-three participants were recruited to take part in Experiment 1, and 23 different participants were recruited to take part in Experiment 2 of this study. In Experiment 1, one participant withdrew after a few blocks of the task because they were feeling dizzy, and thus their data were excluded from further analysis. Two participants were unable to perform the task at the minimum required level of accuracy and were excluded from analysis in Experiment 2. A total of 22 (M = 21.5 years old, SD = 1.79 years old, 10 females) and 21 (M = 22 years old, SD = 3.85 years old, 11 females) participants were thus included in the final samples for Experiment 1 and 2 respectively. All participants provided informed written consent before participation in the study and were compensated for their time (10€/hour), and all procedures were approved by the University College Dublin Human Research Ethics Committee.

Task: Participants performed a motion direction discrimination task adapted from Kiani et al. 2013 [using random dot kinematograms \(RDK\)](#). The experiments consisted of a total of 700 (Exp 1) and 933 trials (Exp 2), divided into blocks of 70 trials each. Most trials in the experiment were two-pulse trials (85.7%) and the remaining fraction were single-pulse trials (14.3%). In single-pulse trials, the second pulse was not presented. Instead, the first pulse was followed by a 0% coherence lead-out period and then the trial terminated. The proportion of single- and two-pulse trials mirror the ones in the original Kiani et al. 2013 [study](#).

Stimuli were presented centrally on a standard monitor, and participants sat at a 50-60cm distance from the monitor. Each trial began with a white fixation cross presented at the centre of the display on black background. Then, a patch of 0% coherence dots appeared in blue colour on the centre of the screen (700 – 1050 ms, with short (700/750ms), medium (900/950ms) or long (1000/1050ms) delays appearing with a probability of 0.7, 0.2 and 0.1, respectively). This was followed by a 200 ms pulse of yellow dots moving at either high or low coherence. After a gap of variable duration (Exp 1: [0, 0.5, 1] s, Exp 2: [0, 0.12, 0.36, 1.08] s) a second 200 ms pulse of either high or low coherence motion was presented. Finally, a patch of blue dots moving at 0% coherence was presented (700 to 1050 ms) in blue. The fixation cross was visible throughout the trial to encourage participants not to follow the dots with their eyes. At the end of the final blue dot sequence, all stimuli disappeared. Trials were separated by a fixed 1s intertrial interval. All gap durations were equally likely, and all possible motion coherence combinations (low-low, low-high, high-low, high-high) and gap durations were randomly permuted throughout the experiment. Three gap durations were originally used in Experiment 1 in order to maximise trial count for each of a small number of gap durations. Having observed that the participants utilised the second pulse less than the first, opposite to previous work (Kiani et al. 2013 [study](#)), we ran a second experiment with shorter gap durations made more prevalent, on the basis that closer temporal contiguity might make it less likely that participants would disengage from accumulation before the second pulse was presented and encourage continued accumulation. The expectations for a long gap to occur differed in our two experiments. Given the percentage of 1-pulse trials, if the second pulse had not yet occurred 500 ms after the 1st pulse the probability of it happening was higher in Experiment 1 ($p = 0.666 [1/3 \cdot 85.7 / (1/3 \cdot 85.7 + 14.3)]$) than in Experiment 2 ($p = 0.6 [1/4 \cdot 85.7 / (1/4 \cdot 85.7 + 14.3)]$). Participants were told they had to respond after stimulus offset, with no time limit. Participants responded by pressing the “C” or “M” keys on a standard keyboard with the left and right thumb respectively. The blue and yellow colours used in the experiment were selected to be physically isoluminant, as measured on the lab monitor, in order to avoid large evoked potentials arising from strong luminance changes.

Staircase procedure: In order to estimate the required dot coherence for participants to perform at 70% accuracy across coherence levels, all participants performed a staircase procedure consisting of 100 trials. All trials consisted of 0% coherence lead-in dots (700 to 1050 ms), a single pulse of evidence (200 ms) and a final sequence of 0% coherence lead-out dots (700 to 1050 ms). The staircase followed a 2-down, 1-up procedure (Levitt 1970), and the starting value was set to 40% for all participants. After two correct responses, the coherence was lowered. The initial step size was set to 6%, and it was halved every second time the staircase went down. This was done to achieve greater precision towards the end of the staircase. The average coherence of the dot motion in the last 40 trials of the staircase procedure were used to estimate the two coherence strengths used in the task. Specifically, high (Exp. 1: $M = 44.73\%$, $SD = 12.64\%$; Exp 2: $M = 33.04\%$, $SD = 13.41\%$) and low (Exp. 1: $M = 26.75\%$, $SD = 7.74\%$; Exp 2: $M = 19.82\%$, $SD = 8.04\%$) coherences were then set to be 25% higher or lower than the staircase-estimated coherence.

To keep performance constant throughout the task, the average accuracy was checked after each block and coherence was dynamically adjusted between blocks. If accuracy was on average lower than 60% or higher than 90% across conditions, coherence was adjusted by increasing or decreasing it by a tenth of the starting value obtained from the staircase procedure. This

precaution was taken to avoid ceiling & floor effects. Most participants' performance was stable throughout the task, and only a few required substantial adjustments of coherence between blocks (see Fig. S1 for individual data).

EEG recording and preprocessing: Continuous EEG was recorded from 128 scalp electrodes using a BioSemi system with a sample rate of 512 Hz. Eye movements were recorded with four additional electro-oculogram (EOG) electrodes, two above and below the left eye and two at the outer canthi of each for vertical (VEOG) and horizontal HEOG) eye. Additionally, we recorded electromyogram (EMG) signals from the thenar eminence using two electrodes on the left and two on the right hand, as a standard procedure, though EMG was not of interest in the current study. All data were analysed in a Matlab R2021a using the EEGLab toolbox (Brunner et al., 2013 [DOI](#); Delorme & Makeig, 2004 [DOI](#)).

Continuous data were high-pass filtered at 0.1Hz, and low-pass filtered at 35Hz using a zero- phase shift Butterworth filter. Then, stimulus-locked epochs were generated from -1.5s to 3s around the onset of the first evidence pulse. Bad channels were defined as those exceeding 50 standard deviation units, and were interpolated. Data were then re-referenced to the common average of all EEG channels. Trials where participants blinked during one of the evidence pulses were excluded from analysis. Eye movements occurring at any other time during the trial were removed from the data using Independent Component Analysis (ICA). Epochs were baseline-corrected relative to the 200 ms before the onset of the first evidence pulse, and channels showing $\pm 150\mu V$ deviations were interpolated on a trial-by-trial basis using the TBT toolbox (Ben-Shachar, 2018 [DOI](#)). If any given epoch had more than 20 bad channels, it was marked for rejection. Further, if any given channel was marked as "bad" on >30% of epochs, it was interpolated across the whole experiment. Finally, we applied a Current Source Density (CSD) transform (second spatial derivative) using the CSD toolbox (Kayser & Tenke, 2006 [DOI](#)) to the preprocessed data to reduce the spatial spread and thereby reduce overlap between topographic foci. Time frequency decomposition was performed using a 7-cycle morlet wavelet transform, extracted in 1-Hz steps in the 1-40Hz range. Alpha and beta band power were obtained by averaging power in the 8-12Hz and 13-30Hz frequency ranges, respectively.

Behavioural analysis

First, to investigate the effect of pulse coherence and gap duration on participants' accuracy, we ran the following logistic regressions on one-pulse (Eq. 1) and two-pulse trials (Eq. 2):

$$\text{Logit}(\text{Accuracy}) \sim \beta_0 + \beta_1 p1coh + [\beta_{0,s} + \beta_{1,s} p1coh] \quad (1)$$

$$\text{Logit}(\text{Accuracy}) \sim \beta_0 + \beta_1 p1coh + \beta_2 p2coh + \beta_3 Gap + \beta_4 p1coh * Gap + \beta_5 p2coh * Gap + [\beta_{0,s}] \quad (2)$$

Where terms between square brackets indicate random effects; s refers to each participant, and indicates that random effects (intercepts, or slopes) were calculated in addition to fixed effects for that particular coefficient. Next, to test whether participants were generally more accurate in two-pulse compared to one-pulse trials we ran the following logistic regression (Eq. 3):

$$\text{Logit}(\text{Accuracy}) \sim \beta_0 + \beta_1 nPulses + \beta_2 p1Coh + \beta_3 nPulses * p1Coh + [\beta_{0,s} + \beta_{1,s} nPulses + \beta_{2,s} p1Coh] \quad (3)$$

This analysis was restricted to two-pulse trials with equal coherences in the two pulses (HH, LL trials only) so that any effect of pulse number would be exclusively related to the presence of additional motion, rather than potentially capturing order or coherence-consistency related effects.

Then, to investigate the effect of pulse order on participants' accuracy, we ran the following regression (Eq. 4):

$$\text{Logit}(\text{Accuracy}) \sim \beta_0 + \beta_1(p1coh + p2coh) + \beta_2(p2coh - p1coh) + [\beta_{0,s} + \beta_{1,s}(p1coh + p2coh) + \beta_{2,s}(p2coh - p1coh)] \quad (4)$$

where the difference coefficient is expected to not be significant if the order of pulses does not influence performance.

In all regressions, coherence values were normalised within participant because they were staircased for each participant individually and varied widely. Where possible, we included all main effects of interest as random effects to control for interindividual variability. These regressions were performed on R using the lme4 toolbox (Bates, 2011 [\[13\]](#)).

Finally, we compared participants' accuracy on two-pulse trials to the expected performance assuming perfect integration, and based on their accuracy in one-pulse trials. Following Kiani et al. 2013 [\[14\]](#), we assumed that decisions are determined by the sign of the sum of evidence in the two pulses. The evidence distribution was inferred from participants' accuracy on single-pulse trials as follows:

$$\begin{aligned} e1 &\sim \Phi^{-1}(P_1, 0, 1) \\ e2 &\sim \Phi^{-1}(P_2, 0, 1) \end{aligned} \quad (5)$$

Where P1 and P2 are grand-average accuracies for high- and low-coherence one-pulse trials.

Φ is the normal cumulative distribution function:

$$\Phi(x, \mu, \sigma) \sim \int_{-\infty}^x N(v, \mu, \sigma) dv \quad (6)$$

and Φ^{-1} is its inverse. $N(v, \mu, \sigma)$ is the normal probability density function with mean ($\mu = 0$) and standard deviation ($\sigma = 1$). The expected accuracy for double-pulse trials was then calculated as follows:

$$P_e \sim 1 - \Phi(0, e1 + e2, \sqrt{2}) \quad (7)$$

ERP analysis

CPP-P1 & CPP-P2 analysis: To investigate how centroparietal responses to the two sequential pulses varied with motion coherence, we extracted EEG activity from a cluster of centroparietal electrodes where activity at 500 ms after pulse 1 was maximal across conditions (see **Fig 2** [\[15\]](#)).

For CPP-P2 visualisation, we removed P1-related activity by subtracting the traces from the longest gap conditions [-0.2 to 1.4s with respect to P1], from all traces. This resulted in an average EEG activity of 0 (see *Fig. S4*) preceding the second pulse, in all conditions, and rendered the key CPP-P2 findings more clearly visible by removing P1-induced EEG drifts. Note that this does not affect the CPP-P2 activity in the long gap conditions, as the subtraction alters data up to the end of P2 but not further (see *Fig. S4*). We report the average across all non-zero gap conditions in the main manuscript (**Fig. 2** [\[15\]](#)), as well as the data sorted by gap in the supplementary material (*Fig. S3, S4*). We also used the so-corrected activity to compare the relative amplitudes of CPP-P1 and CPP-P2 across all non-zero gap conditions. For this, we first computed the average amplitude 0.4 to 0.6s after P1 (in the raw data, *CPP-P1amp*) and P2 (in the corrected data, *CPP-P2amp*), for each gap separately, and averaged across all non-zero gap conditions. Then, we computed the percentage of CPP-P1 amplitude reached by CPP-P2 as $\text{CPP-P2amp}/\text{CPP-P1amp} * 100$.

To investigate the impact of dot motion coherence on both CPP-P1 and CPP-P2 amplitudes (**Fig. 3**), we ran a series of time-resolved regressions:

$$CPP_t \sim p1Coh + p2Coh \quad (8)$$

Where t indicates time from P1 for CPP-P1 analysis, and time after P2 for CPP-P2 analysis. For this analysis, we aligned all gap conditions to P1/P2 onset respectively and ran the regressions for all trials pooled, for each subject separately. For CPP-P2, data were additionally baselined using 100ms before P2 onset to minimise the impact of gap duration on average amplitudes. All EEG data were low-pass filtered (6Hz) prior to running the regression to reduce the impact of high frequency noise on these single-trial analyses.

Next, to investigate how trial-to-trial variability in attention and evidence accumulation influenced performance, we ran a series of time-resolved logistic regressions.

First, we investigated whether single-trial variability in centroparietal responses modulated the weight of pulses on choice. The interaction terms between pulse coherence and CPP amplitude in the following regression (Eq. 9) thus test whether pulses which evoked a particularly high centroparietal response were also associated with that pulse having a greater weight on choice:

$$\text{Logit}(P(\text{Choice} = \text{right})) \sim p1Coh + p2Coh + p1Coh * CPP_t + p2Coh * CPP_t \quad (9)$$

Where t indicates time after P1 for CPP-P1 analysis, and time after P2 for CPP-P2 analysis.

Second, we ran an analogous analysis to investigate whether fluctuations in evoked occipital alpha desynchronisation also modulated the weight of each pulse on choice (Eq. 10):

$$\text{Logit}(P(\text{Choice} = \text{right})) \sim p1Coh + p2Coh + p1Coh * -\text{Alpha}_t + p2 * -\text{Alpha}_t \quad (10)$$

Where t indicates time after P1 for alpha-P1 analysis, and time after P2 for alpha-P2 analysis.

All regressors and EEG signals were z-scored across trials by timepoint. For alpha power analysis, we took the *negative* alpha power across a range of bilateral occipital electrodes (see *highlighted* electrodes in **Fig. 5**), so that positive coefficient values indicate that higher attention (i.e. lower alpha power) increase weight on choice or CPP amplitude. For the CPP analysis, we used the signal averaged over a cluster of centroparietal electrodes for CPP (**Fig. 2B**). Regressions were run for each gap condition separately for times between [0-1s] after each pulse, and the resulting coefficients were aligned to P1 and P2 and averaged across conditions.

EEG statistics: EEG evoked potentials and regression results were statistically evaluated against zero using cluster-based permutations using the FieldTrip toolbox (Maris & Oostenveld, 2007; Oostenveld et al., 2011). All tests were two-tailed, cluster-corrected ($p < 0.05$), and 10000 permutations were run. We additionally compared the magnitude of the occipital alpha level in the longest gap across the two experiments using an independent samples t-test in Matlab.

Model fitting & simulations: In order to account for participants' behaviour, we fit a simple diffusion model to the grand-averaged accuracy in the 4 conditions with three free parameters: bound (b) and two drift rates ($d1$, $d2$) corresponding to the high and low coherence conditions. The accumulator model followed this basic equation:

$$y_t \sim y_{t-1} + d + N_{0,\sigma} \quad (11)$$

Where y_t is the accumulator output or decision variable (DV) for any time t within a pulse (where $d=d1$ or $d = d2$ depending on coherence). Gaussian noise, $N(0,\sigma)$, was added in order to represent noisy sensory evidence through the trial. During the gap, the accumulator was set to stay stable at its final within-pulse value so that y_t was set to equal y_{t-1} for any time t outside of pulses during which incoherent motion was presented and dots were blue. Given the behavioural observation that overall accuracy did not decrease in longer gap durations, and the fact that evidence pulses were clearly indicated by a change in the dots colour, the DV was paused and not allowed to further accumulate the noise during the 0% coherence motion gap separating the two pulses. When the DV crossed a bound ' b ', evidence accumulation terminated. The accumulator assumed no noise or leak during the gap, and in any trials where the bound was not crossed accumulation continued upon presentation of the next pulse. Note that this one-dimensional, continuous, bounded accumulator represents the overall DV corresponding to the level of motor preparation. As we explain below, the CPP was simulated by extracting the single-pulse components of this model.

We fit the model by minimising G^2 with the *particleswarm* function in the Global Optimisation Toolbox for Matlab. G^2 is a goodness of fit measure that in this study quantified how much predicted accuracies differed from observed ones in each condition.

Computational models were fit to the two experiments separately simulating 10000 trials for each condition, and the best fitting parameters were then used to simulate CPP traces. Akaike Information Criterion (AIC) and Bayes Information Criterion (BIC) were computed to assess the goodness of fit as follows:

$$AIC = G^2 + 2 * k \quad (12)$$

$$BIC = G^2 + k * \log(n) \quad (13)$$

Where k indicates the number of free parameters for a given model, and n represents the number of trials used to fit the model.

Neural data simulation

We simulated MBL activity by taking the evolving value of the simulated decision variable throughout the trial. If a bound was hit, the DV was set to stay stable in order to account for the sustained nature of MBL signals our empirical data as well as previous work in delayed response tasks (Murphy et al., 2021 [DOI](#); Parés-Pujolràs et al., 2025 [DOI](#); Twomey et al., 2016 [DOI](#)).

In turn, following previous work, we simulated the CPP by taking the absolute value of the underlying, signed accumulator traces unfolding during each individual pulse, produced by the best fitting parameters of our model (Afacan-Seref et al., 2018 [DOI](#); Kelly et al., 2021 [DOI](#)). Taking the absolute value implicitly assumes that the differential evidence accumulator is implemented via two separate neural populations that encode values respectively favouring one or the other alternative. If the simulated decision variable hit a bound, the accumulator traces were set to fall back down to zero within 200 ms after hitting the bound, based on past observations of post-peak CPP dynamics (Steinemann et al. 2018 [DOI](#)). In the model shown in the main text, if the overall DV did not hit a bound by the end of the first pulse, it was assumed that the cumulative evidence level attained thus far was sustained in the overall DV at the motor level, but that the intermediate accumulator fell back to its zero baseline level, and restarted accumulation upon the presentation of the second pulse, which at the motor level added to the already sustained level. Variants of this scheme were also explored, as explained in Fig. S7, S8. In order to qualitatively compare how well various decay parameters matched the empirical CPP dynamics, we computed the relative CPP-P2/CPP-P1 percentage amplitude predicted by various models by extracting the average evoked amplitude of the simulated CPP [0-0.2s] for each pulse separately, with respect to the pre-pulse

baseline amplitude. This 0.2s interval is analogous to the time used for analysis of the empirical data (see above), but is set to an earlier time period because our simulations did not incorporate any sensory delay, and therefore ramping activity started instantly after stimulus onset.

Supplementary material

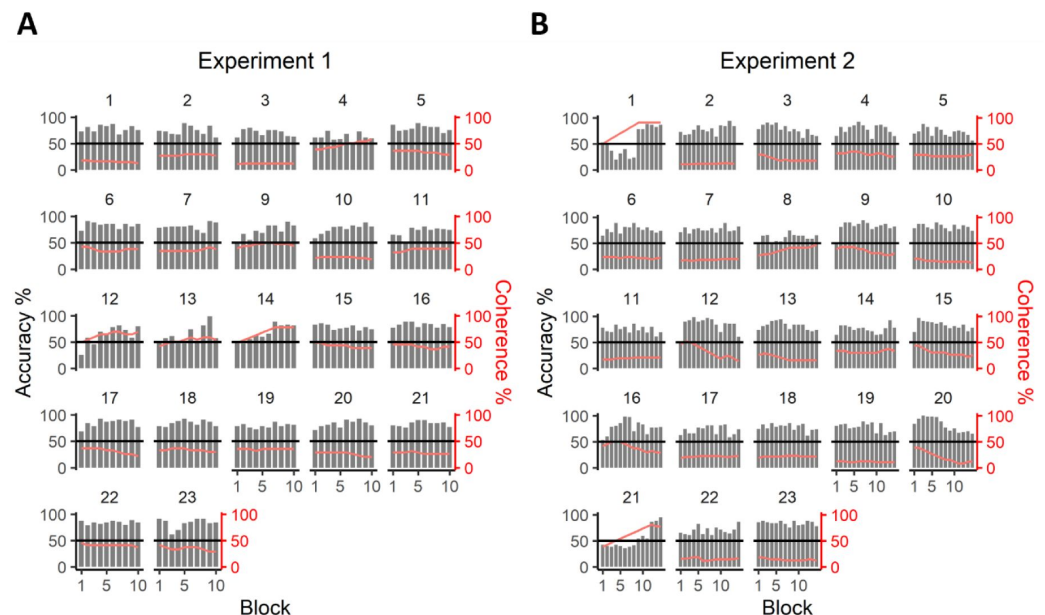


Figure S1.

Individual participants' accuracy (grey bars) and average coherence (red line) for each block in Experiments 1 (A) and 2 (B).

On average, most participants performed close to 70% accuracy and clearly above chance (solid black line), and only required minimal coherence adjustments between blocks. The data of participants 1 and 21 in Experiment 2 were excluded from further analysis due to poor task performance in a majority of blocks.

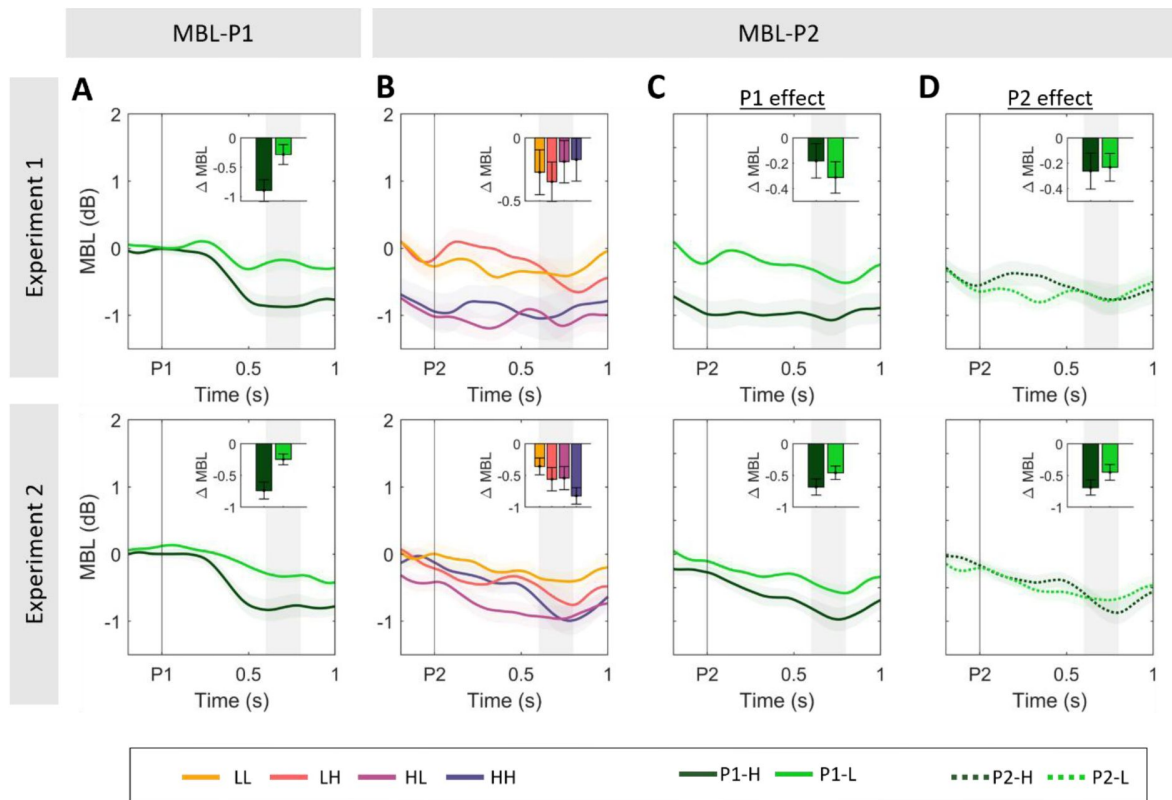


Figure S2.

Sensitivity to evidence strength in the two pulses in motor beta lateralisation signals.

A. Grand-averaged ($\pm$ SEM) motor beta lateralisation [Contra-Ipsilateral hemispheres with respect to correct response hand] after pulse 1 (MBL-P1), averaged across all gap conditions and across all trials regardless of accuracy. Motor preparation started shortly after the first pulse, and was sustained at a coherence-dependent level. **B.** Grand-averaged MBL after pulse 2 (MBL-P2), averaged over all non-zero gaps in the two experiments. **C.** Effect of P1 coherence on MBL after P2. Traces illustrate the grand-averaged ($\pm$ SEM) aligned to pulse 2 (MBL-P2), averaged over all non-zero gaps and across P2 coherence. **D.** Effect of P2 coherence on MBL after P2. Traces illustrate the grand-averaged ($\pm$ SEM) aligned to pulse 2 (MBL-P2), averaged over all non-zero gaps and across P1 coherence. In panels **A-D**, inset bar graphs illustrate the MBL excursion as the difference between activity at [0.6-0.8s, shaded area] after pulse onset, minus activity at baseline [-0.2-0s].

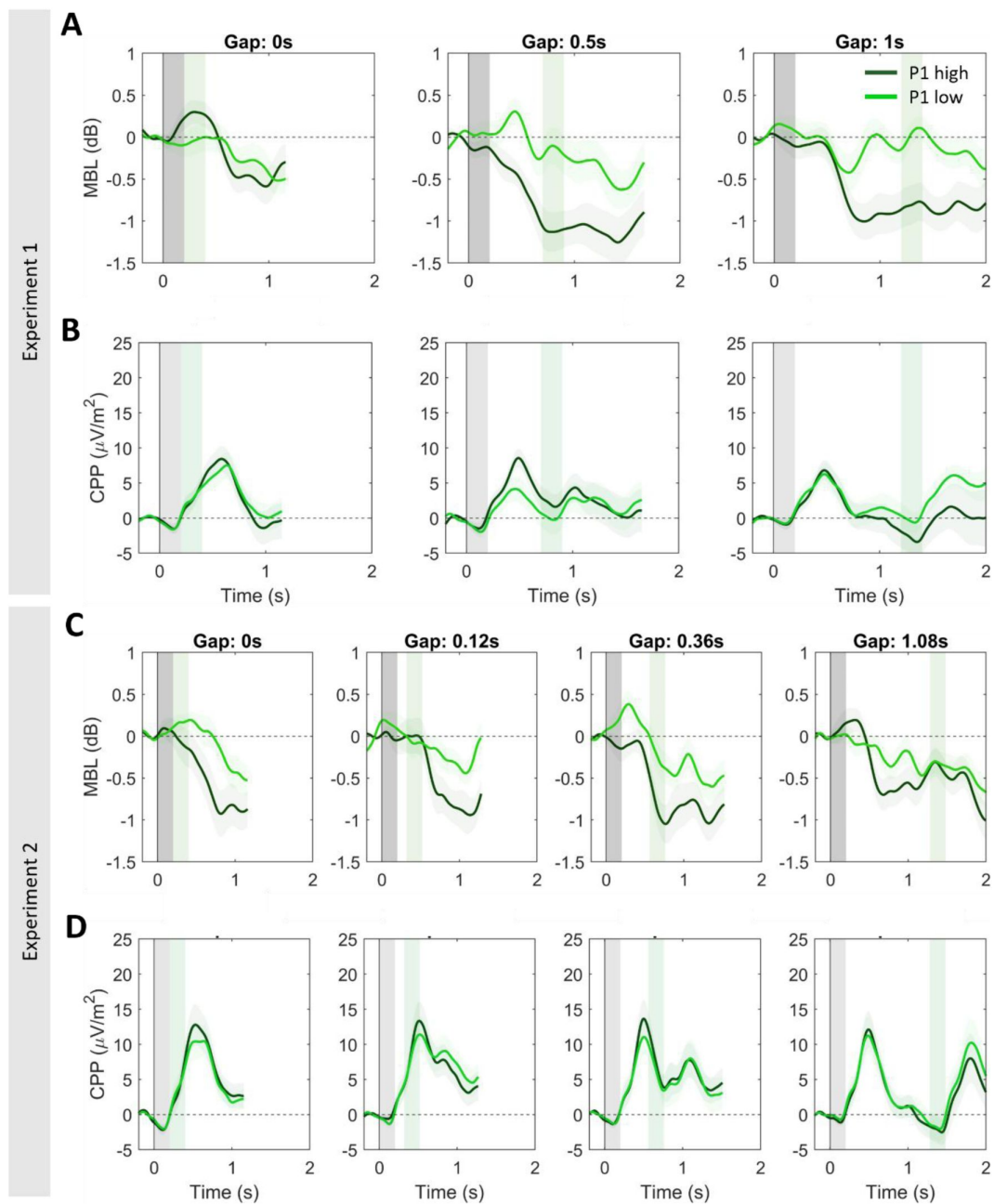


Figure S3.

A,C. Motor beta lateralisation (MBL) for each gap condition, sorted by P1 coherence in experiments 1 (A) and 2 (C). Lateralisation is computed as a function of actual dot motion [Contra-Ipsilateral hemispheres with respect to correct response hand], sorted by gap duration and P1 coherence pooled across correct & error trials. **B,D.** Centroparietal activity for each gap condition, and sorted P1 coherence in experiments 1 (B) and 2 (D).

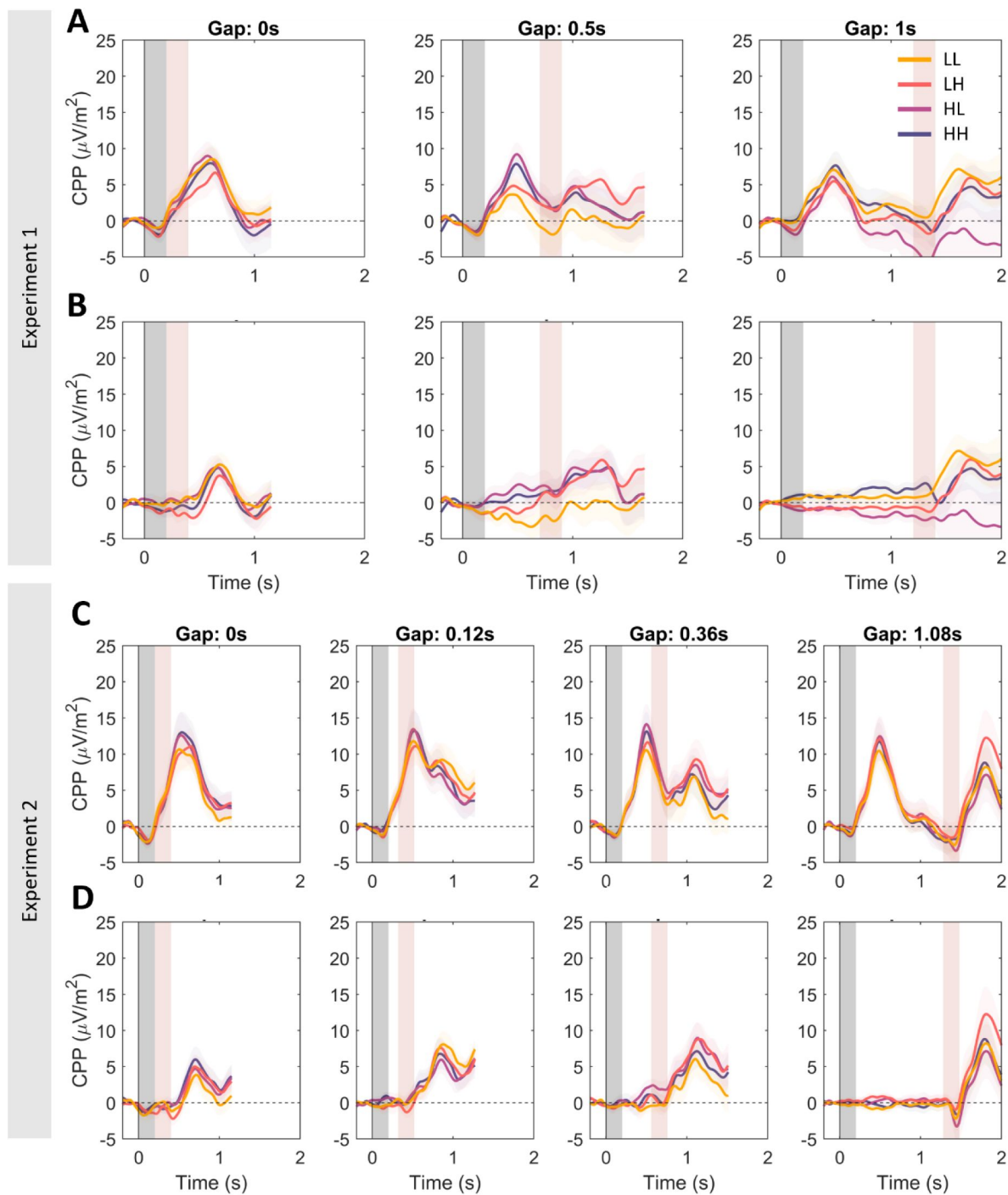


Figure S4.

A,C. Centroparietal activity sorted by gap, P1 and P2 coherence in Experiments 1 and 2. **B,D.** Centroparietal activity after removing CPP-P1 activity. For each condition, the average of long-gap trials with the same P1 coherence was removed. That is, the average of the long gap HH & HL conditions was subtracted from HH and HL traces. Similarly, the average activity of the long gap LL & LH conditions was subtracted from LL and LH traces. This removes overlapping CPP-P1 potentials, and isolates centroparietal activity related to P2. Subtractions were performed between [-0.2 to 1.4s] from P1 presentation. Thus, CPP-P2 potentials in the long gap condition remain unaltered.

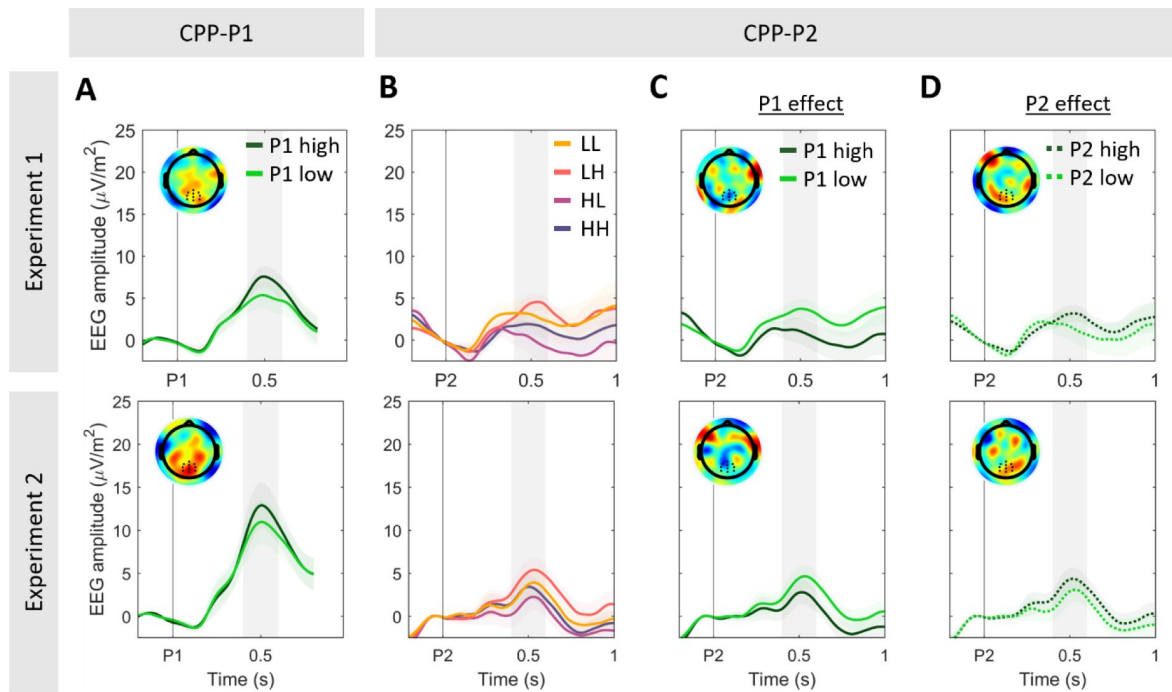


Figure S5.

Replication of results in Fig. 3, now performing the analysis on CPP-P2 data (B- D) without removing CPP-P1 activity.

Note that baseline activity shows a drift consistent with the overlapping potentials from the preceding pulse, which is corrected for when removing CPP-P1 activity (cf. Fig. 3). Note also that the same patterns are qualitatively present in the uncorrected data. Panels C-D illustrate conditions pooled by P1 or P2 coherence, respectively, to highlight the different direction of the effect.

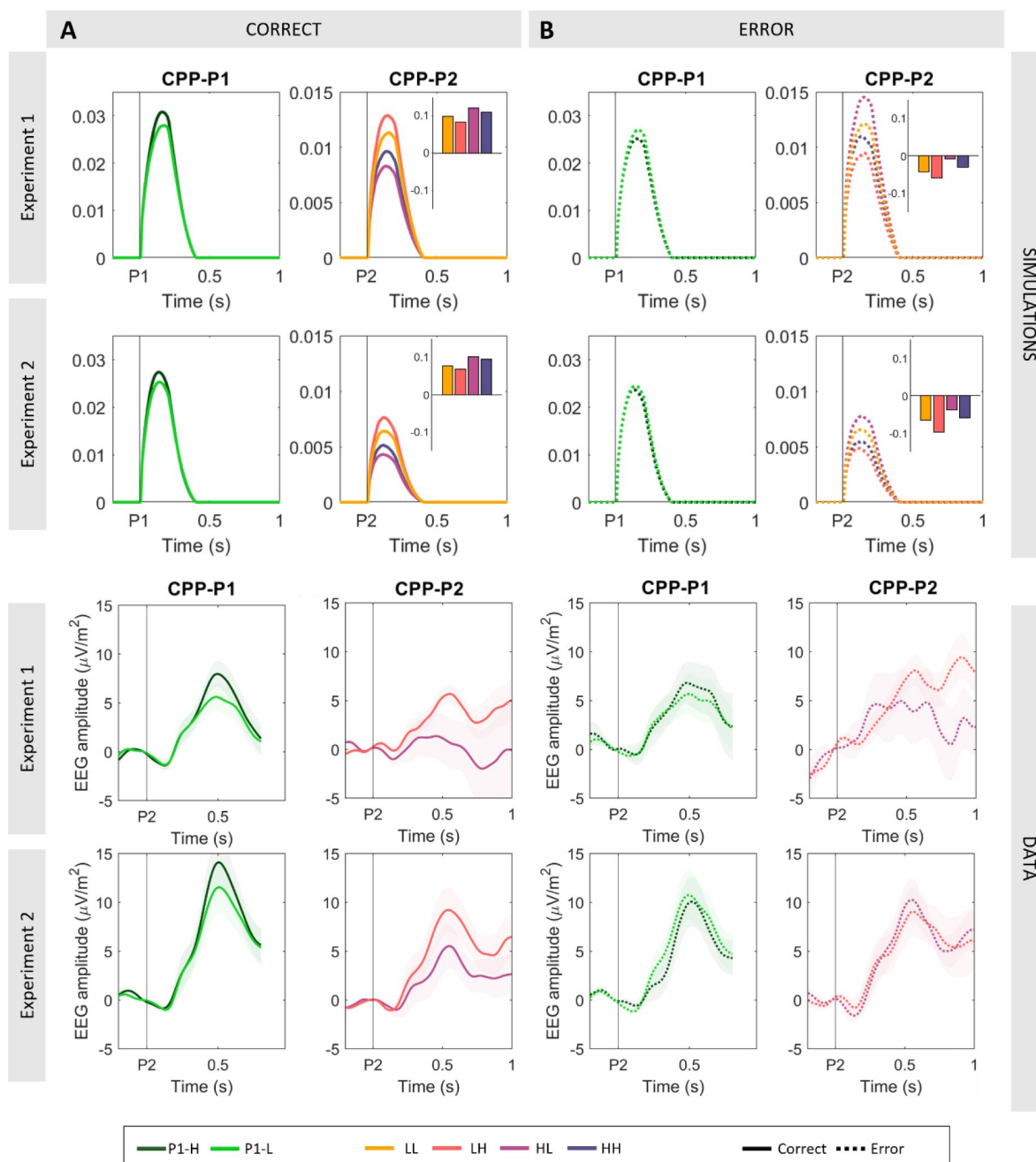


Figure S6.

Model-predicted (A-B) and observed (C-D) CPP-P1 and -P2, in correct vs.

incorrect responses, for Experiments 1 (top) and 2 (bottom). Bar graph insets in A-B indicate the state of the simulated accumulator at the end of pulse 1 in those trials where the bound was not hit. Positive values indicate that the accumulator was leaning towards the correct answer, whereas negative values indicate that it was leaning towards an error. The model predicts stronger P1 coherence effects in correct compared to error trials (A-B, left), with this effect also being observed in empirical data across both experiments (C-D, left). For the sake of clarity, we do not plot LL and HH conditions in the empirical CPP-P2 data because no clear amplitude differences are predicted by the model in correct vs. error trials, and the trial count in the HH condition was insufficient in most participants. Further, the model predicted a strong interaction in CPP-P2 amplitudes in the HH vs. HL condition. While correct trials (which were the majority of trials in our task and the model) showed higher amplitudes at CPP-P2 for HL compared to the LH condition (A, right), consistent with the effect of P1 on CPP-P1 amplitudes reported in [Fig 3](#), this effect reversed in incorrect trials (B, right). This interaction was also clearly visible in Experiment 2, although slightly less prominent in Experiment 1. Note however that Experiment 1 had fewer overall trials, and power to detect differences is likely to be insufficient after splitting conditions by accuracy.

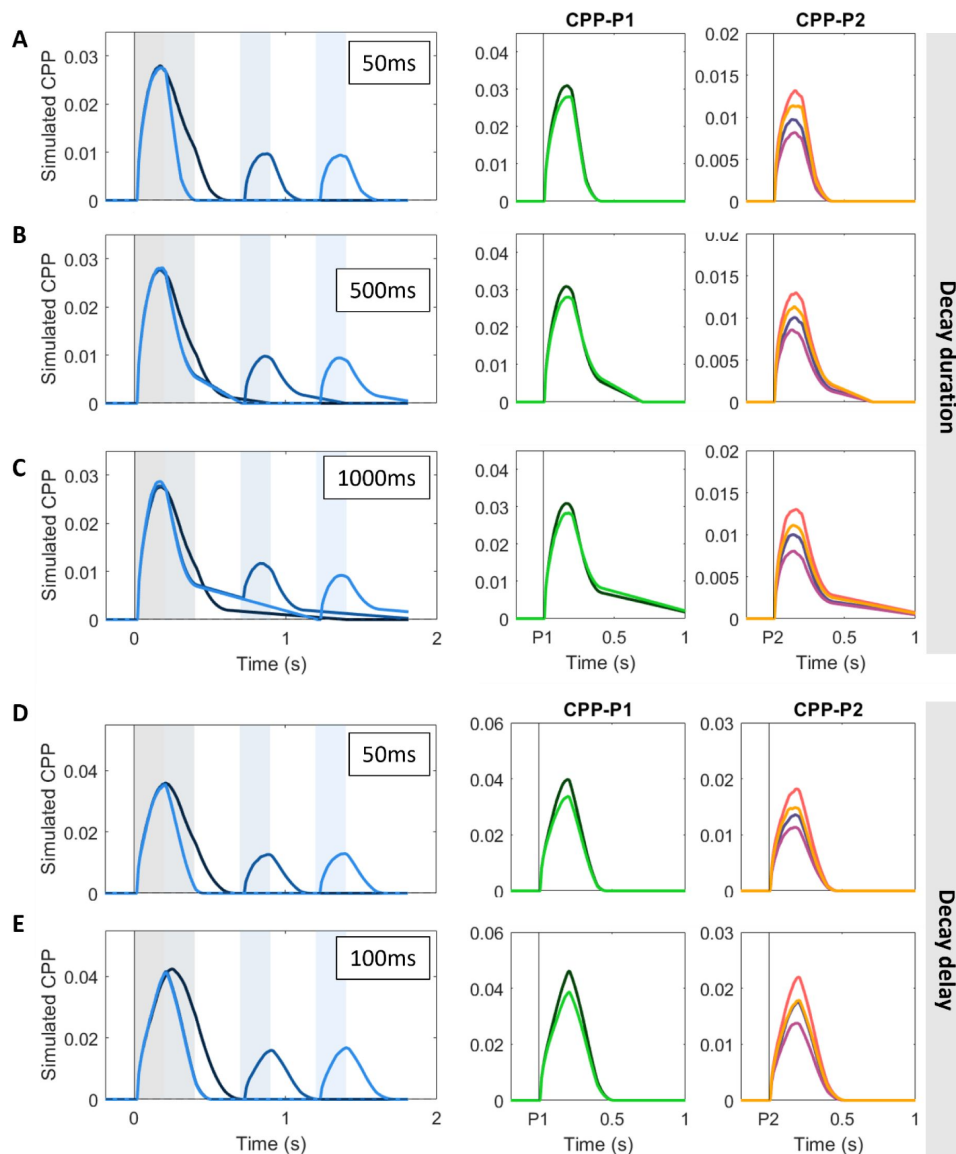


Figure S7.

Simulations (N=1000) illustrating the effect of various signal decay time (A-C) and decay delay (D-E) assumptions on the predicted CPP activity.

Simulations are based on the fitted model in the main manuscript and for Exp. 1 only for illustration purposes. Given that it's unclear whether the CPP fall-down to zero when evidence is paused follows the same dynamics as its fall-down when it hits a bound (where studies with immediate behavioural responses suggest that it reaches zero within approximately 200ms; Steinemann et al. 2018), we simulated its CPP activity under three different assumptions about the duration of its decay upon evidence interruption (i.e. when blue dots appear onscreen after a pulse). We simulated the CPP assuming the signal takes 100 ms (A), 500 ms (B) or 1000 ms (C) to go back to baseline at pulse offset when a bound had not been reached, retaining a 200 ms fall-down upon bound-crossing. Longer signal decay times for trials where no bound was hit result in a protracted ramp down to baseline. In turn, it is unclear whether any post-decisional evidence accumulation is to be allowed to simulate the CPP in this context, given that previous studies have observed varying delays in signal decay after response (Steinemann et al. 2018; Afacan- Seref et al., 2011). In the manuscript, we assumed no delay between bound crossing and signal decay. Here, we simulated the CPP allowing it to continue building up 50ms (D) or 100ms (E) after bound crossing, while setting it to immediately decay upon blue dot offset. Allowing some post-decisional build-up resulted in higher average CPP amplitudes, and it also better recapitulated some features of the data, including a protracted build-up in the zero-gap condition compared to all others.

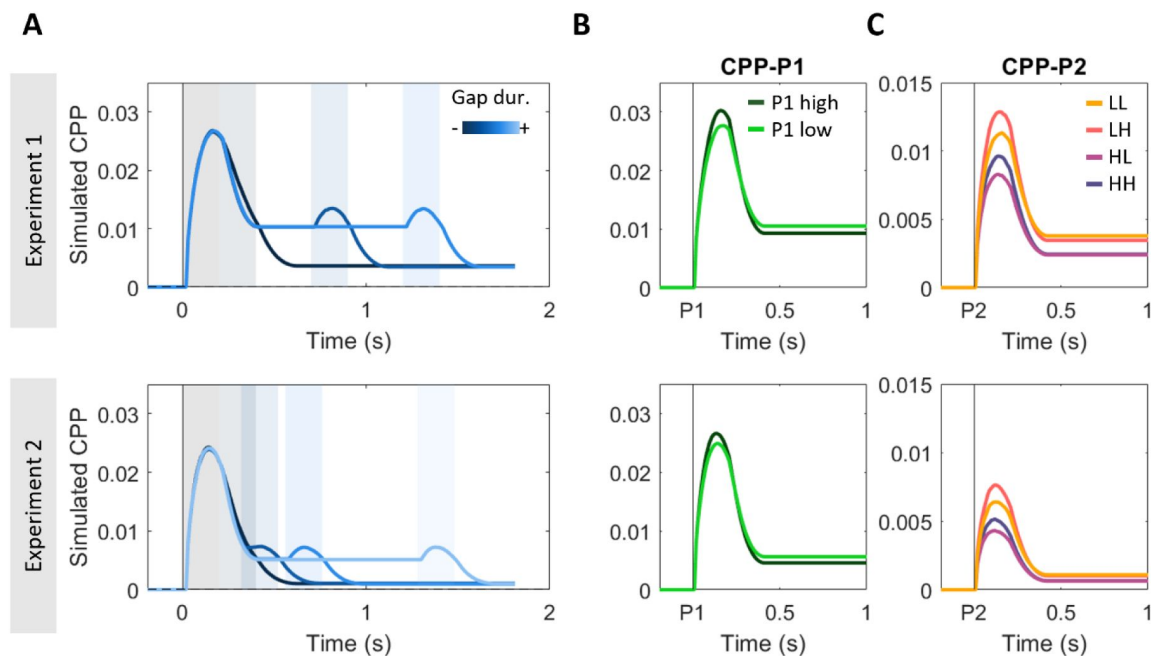


Figure S8.

Simulated CPPs using the same computational model as in Fig. 5, but under the assumption that the signal was only reset to baseline if a bound was hit.

If the bound was not hit, the accumulator, like the downstream DV at the motor level, was set to stay stable at the final value reached by the end of pulse 1 with no decay and no added noise for the whole duration of the gap. Given that we do not model reaction times, we assumed no transmission delay between events on the screen and their impact on the accumulator. In this model, the partial fall down of the signal during the gap was driven by those trials that hit a bound following the first pulse. Conversely, the fact that it did not fall all the way to 0 on average during the gap is driven by those trials where the DV did not hit a bound during processing of pulse 1 and thus sustained through the gap. Accumulation resumed from that sustained level upon presentation of a second pulse, and the fact that on average CPP-P2 traces reach lower amplitudes than CPP-P1 evoked activities is explained by the fact that trials that were terminated during the first pulse did not engage in accumulation again, and therefore took value 0 in our model. Note that this model provides a qualitatively worse fit to the empirical CPP traces during the gap and CPP-P2 trials. In particular, the magnitude of the CPP-P2 potentials was predicted to be significantly smaller (Exp. 1: CPP-P2 = 19% of CPP-P1 peak amplitude; Exp. 2: CPP-P2 = 13% of CPP-P1 peak amplitude) than observed on the grand-averaged data (CPP-P2 = approximately 50% of CPP-P1 in both experiments; see Fig. 2).

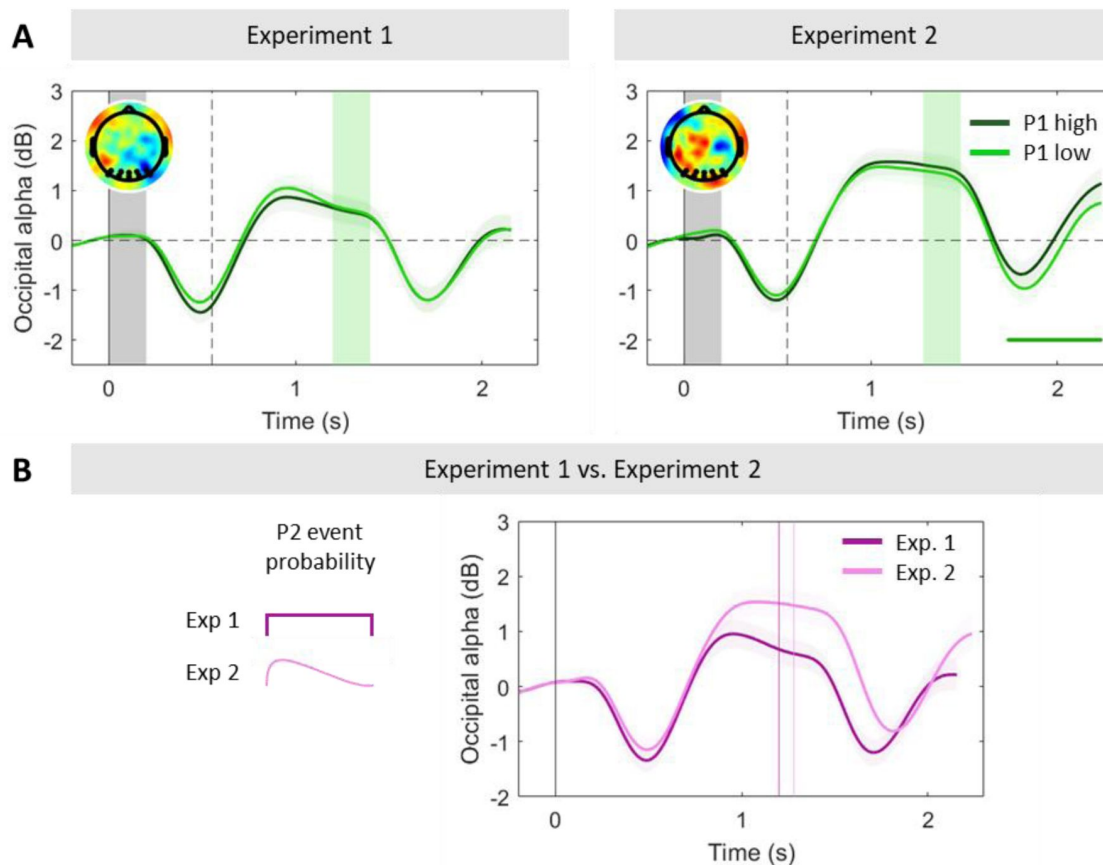


Figure S9.

Occipital alpha power tracks dynamics of attentional engagement and varies with temporal expectations.

A. Grand-averaged ($\pm$ SEM) occipital alpha power in the long gap conditions sorted by P1 coherence, again baseline-corrected before the onset of the first pulse. At evidence offset, alpha steeply increased, and, interestingly, attained and maintained a higher level during the longer gaps than the baseline before the first pulse (two-tailed cluster permutation test, $p < 0.05$ in both experiments; see **Fig. 2A**). One possible interpretation is that this reflects attentional disengagement from the decision occurring after early decision termination. However, if this was the case, we would expect this effect to be dependent on pulse 1 coherence, as it is more likely for a DV to have reached a bound after a high-coherence pulse (**Fig. 4B**), and we observed no such coherence-dependence in alpha power during the gap. Topographies illustrate the difference in alpha power between P1-high and P1-low coherence trials [-0.2 to 0s] before P2 onset. **B.** Occipital alpha power in the longest gap condition in the two experiments. In Experiment 2, where the event distribution was skewed towards short gap durations between the first and second pulses, participants' occipital alpha was significantly higher (indicating lower attention) than in Experiment 1 ($*p < 0.05$; two-sample t-test for the difference in alpha power between occipital activity [-0.1 to 0s] before P2 and the baseline [-0.2 to 0s] before P1. Alpha power has been linked to temporal expectation encoding (Rohenkohl & Nobre, 2011), as well as being a well-known index of attentional engagement (Thut et al., 2006). Thus, this difference is interesting in light of the difference in gap durations in our two experiments, and hence temporal expectations. In particular, in those trials where no second pulse had been presented after a 0.5s gap, the probability of the second pulse yet appearing was 0.67 in Experiment 1, compared to 0.6 in Experiment 2. Although another difference between experiments was that early terminations within pulse 1 occurred more often in Experiment 2, as predicted by our model (see **Fig. 4B**), the probability of early terminations is unlikely to be the key influence given the above finding that alpha power during the gap was not dependent on P1 coherence (**A**). Rather than reflecting switches between engagement and disengagement at the decision level, then, it may reflect a gating function at the sensory processing level that operates independent of whether early bound-crossings have happened, yet is modulated between contexts depending on the likelihood of a second pulse still to come. The dotted box indicates the test period. In all panels, shaded areas indicate the timing of evidence pulses for the relevant gap durations. *Markers along the bottom of panel A indicate significant clusters where alpha power differed from zero (two-tailed cluster-based permutation test, $p < 0.05$). In panel B, they indicate significant clusters where alpha power differed in P1-high vs. P1-low coherence trials.

		b	Free parameters			Goodness of fit		
			D1	D2	Dvar_P2	AIC	BIC	G ²
Exp. 1	Model 1	0.047	0.097	0.177		6.41	20.82	0.41
	Model 2	0.055	0.091	0.181	0.152	8.34	27.55	0.34
Exp 2	Model 1	0.037	0.117	0.205		6.82	22.50	0.82
	Model 2	0.039	0.122	0.211	0.237	8.59	29.36	0.59

Figure S10.

Model comparison investigating the effect of drift rate variability at the second pulse.

In our main analysis, we initially fit a model with two different drift rates for low (d1) and high (d2) coherence pulses: **Model 1** (**k = 3, b, d1, d2**). Here, we explored whether increased temporal uncertainty about whether and when a second pulse might appear could have led to more variable attentional states before its appearance, and as a consequence more variable drift rates. To test this hypothesis, we extended the model with one additional free parameter introducing trial-by-trial variability to the drift rates applied to pulse 2: **Model 2** (**k = 4: b, d_low, d_high, dvar_p2**). This model resulted in the lowest overall error as measured by G², but the additional complexity was not warranted according to AIC/BIC metrics.

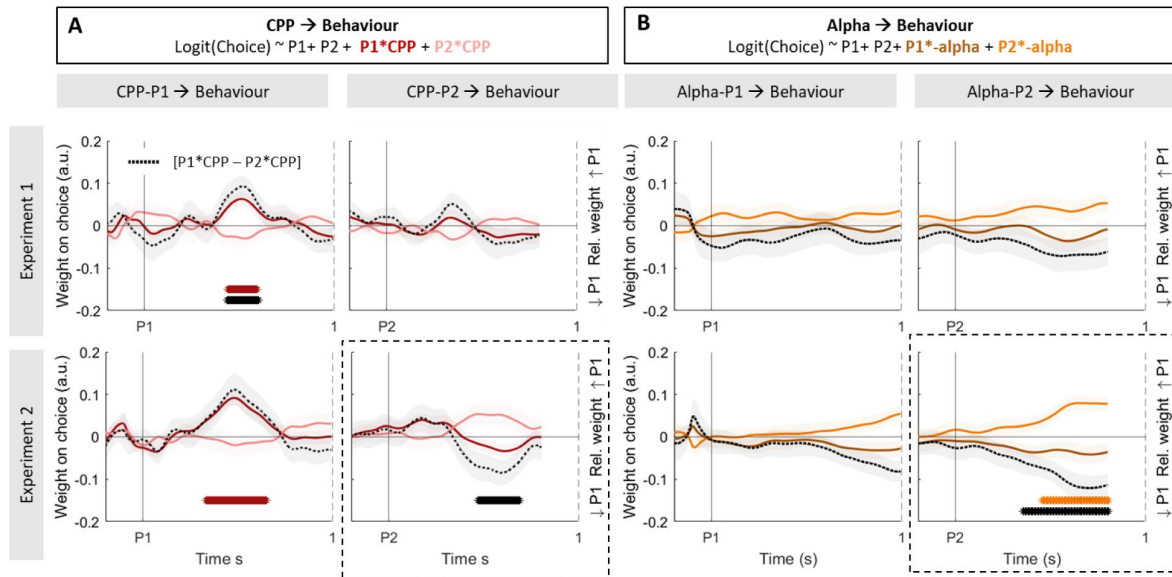


Figure S11.

Weight on choice modulations by single-trial fluctuations in neural evidence accumulation and attention markers.

In this analysis, we sought to investigate whether single-trial fluctuations in the neural correlates of evidence accumulation and attention were linked to variability in choice behaviour. To do so, we exploited the trial-by-trial variability in the neural signals. **A.** Standardized regression coefficients ($\pm$ SEM) resulting from Eq. 9, illustrating the effect of CPP-P1 (*left*) and CPP-P2 (*right*) variability on choice, averaged across all gap conditions. The magnitude of CPP-P1 modulated the impact of the first pulse on choices in both experiments while the magnitude of CPP-P2 only modulated the relative weight of P2 in Experiment 2. In our task, variability in the CPP may be related to trial-by-trial variability in motion energy, as well as random noise. Since the weight on choice of P1 and P2 may trade-off with one another due to a bound being set - i.e. on trials where P1 has a strong influence, P2 is likely to have a weaker one - we also computed the difference between the two modulatory terms ($[P1_{coh} * CPP - P2_{coh} * CPP]$, for CPP-P1 and CPP-P2 respectively). This difference index effectively quantifies whether centroparietal responses to either pulse modulate the relative weight on choice of each of the pulses. A positive index indicates a relative increase of P1 weight on choice, whereas a negative index indicates a relative increase of P2 weight on choice. Black dashed lines correspond to the difference in regression coefficients for the $[P1_{coh} * CPP - P2_{coh} * CPP]$ modulations]. **B.** Standardized regression coefficients ($\pm$ SEM) resulting from Eq. 10 for alpha aligned to pulse 1 (*left*) or pulse 2 (*right*), averaged across all gap conditions. The magnitude of occipital phasic alpha responses (baselined before P1) did not modulate the weight on choice of P1, and after P2 did modulate its weight on choice in Experiment 2 only. Black dashed lines correspond to the difference in regression coefficients for the $P1 * \alpha - P2 * \alpha$ modulations, which effectively indicate a change in the relative weight for P1 and P2. *Asterisks along the bottom of the panels indicate significant ($p < 0.05$) effects in the test period [0.2-0.8s post event].

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Additional information

Author contributions

Conceptualisation: E.P-P, & S.P.K; software: E.P-P; data curation & analysis: E.P-P.; resources: S.P.K.; writing of original draft: E.P-P.; review & editing: E.P-P, R.G.O., A.C.G, & S.P.K.; supervision: S.P.K.

Data & code

Data and analysis code will be made publicly available upon publication.

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Reviewer #1 (Public review):

Summary:

This paper aims to characterise the physiological and computational underpinnings of the accumulation of intermittent glimpses of sensory evidence.

Strengths:

- (1) Elegant combination of electroencephalography and computational modelling.
- (2) The authors describe results of two separate experiments, with very similar results, in effect providing an internal replication.
- (3) Innovative task design, including different gap durations.

Weaknesses:

- (1) The authors introduce the CPP as tracking an intermediary (motor-independent) evidence integration process, and the MBL as motor preparation that maintains a sustained representation of the decision variable. It would help if the authors could more directly and quantitatively assess whether their current data are in line with this. That is, do these signals exhibit key features of evidence accumulation (slope proportional to evidence strength, terminating at a common amplitude that reflects the bound)? Additionally, plotting these

signals report locked (to the button press) would help here. What do the results mean for the narrative of this paper?

(2) The novelty of this work lies partly in the aim to characterize how the CPP and MBL interact (page 5, line 3-5). However, this analysis seems to be missing. E.g., at the single-trial level, do relatively strong CPP pulses predict faster/larger MBL? The simulations in Figure 5 are interesting, but more could be done with the measured physiology.

(3) The focus on CPP and MBL is hypothesis-driven but also narrow. Since we know only a little about the physiology during this "gaps" task, have the authors considered computing TFRs from different sensor groupings (perhaps in a supplementary figure?).

(4) The idea of a potential bound crossing during P1 is elegant, albeit a little simplistic. I wonder if the authors could more directly show a physiological signature of this. For example, by focusing on the MBL or occipital alpha split by the LL, LH, HL and HH conditions, and showing this pulse- as well as report-locked. Related, a primacy effect can also be achieved by modelling (i) self-excitation of the current one-dimensional accumulator, or (ii) two competing accumulators that produce winner-take-all dynamics. Is it possible to distinguish between these models, either with formal model comparison or with diagnostic physiological signatures?

(5) The way the authors specify the random effects of the structure of their mixed linear models should be specified in more detail. Now, they write: "Where possible, we included all main effects of interest as random effects to control for interindividual variability." This sounds as if they started with a model with a full random effect structure and dropped random components when the model would not converge. This might not be sufficiently principled, as random components could be dropped in many different orders and would affect the results. Do all main results hold when using classical random effects statistics on subject-wise regression coefficients?

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Reviewer #2 (Public review):

Summary:

This manuscript examines decision-making in a context where the information for the decision is not continuous, but separated by a short temporal gap. The authors use a standard motion direction discrimination task over two discrete dot motion pulses (but unlike previous experiments, fill the gaps in evidence with 0-coherence random dot motion of differently coloured dots). Previous studies using this task (Kiani et al., 2013; Tohidi-Moghaddam et al., 2019; Azizi et al., 2021; 2023) or other discrete sample stimuli (Cheadle et al., 2014; Wyart et al., 2015; Golmohamadian et al., 2025) have shown decision-makers to integrate evidence from multiple samples (although with some flexible weighting on each sample). In this experiment, decision-makers tended not to use the second motion pulse for their decision. This allows the separation of neural signatures of momentary decision-evidence samples from the accumulated decision-evidence. In this context, classic electroencephalography signatures of accumulated decision-evidence (central-parietal positivity) are shown to reflect the momentary decision-evidence samples.

Strengths:

The authors present an excellent analysis of the data in support of their findings. In terms of proportion correct, participants show poorer performance than predicted if assuming both evidence samples were integrated perfectly. A regression analysis suggested a weaker weight on the second pulse, and in line with this, the authors show an effect of the order of pulse

strength that is reversed compared to previous studies: A stronger second pulse resulted in worse performance than a stronger first pulse (this is in line with the visual condition reported in Golmohamadian et al., 2025). The authors also show smaller changes in electrophysiological signatures of decision-making (central parietal positivity and lateralised motor beta power) in response to the second pulse. The authors describe these findings with a computational model which allows for early decision-commitment, meaning the second pulse is ignored on the majority of trials. The model-predicted electrophysiological components describe the data well. In particular, this analysis of model-predicted electrophysiology is impressive in providing simple and clear predictions for understanding the data.

Weaknesses:

Some readers may be left questioning why behaviour in this experiment is so different from previous experiments, which use almost exactly the same design (Kiani et al., 2013; Tohidi-Moghaddam et al., 2019; Azizi et al., 2021; 2023). The authors suggest this may be due to the staircase procedure used to calibrate the coherence of (single-pulse) dot motion stimuli for individuals at the start of the experiment. But it remains unclear why overall performance in this experiment is so bad. Participants achieved ~85% correct following 400 ms of 33 - 45% coherent motion. In previous work, performance was ~90% correct following 240ms of 12.8% coherent motion. It seems odd that adding the 0% coherent motion in the temporal gaps would impair performance so greatly, given it was clearly colour-coded. There is a lack of detail about the stimulus presentation parameters to understand whether visual processing explains the declined performance, or if there is a more cognitive/motivational explanation.

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